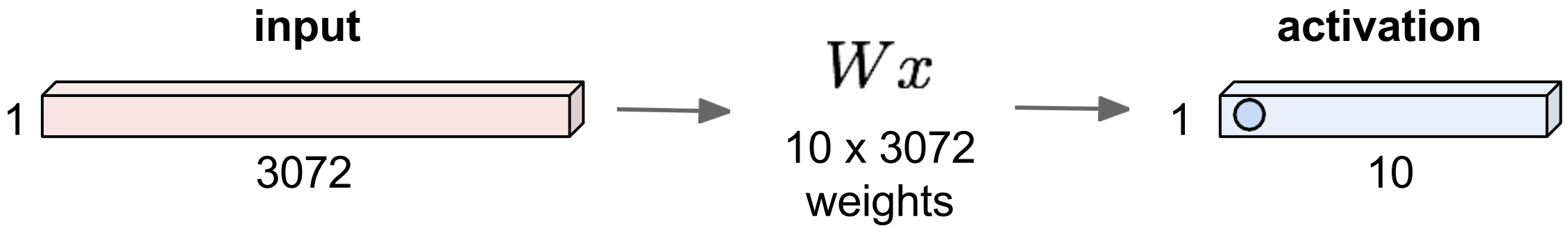
卷积神经网络

Convolutional Neural Networks

全连接层

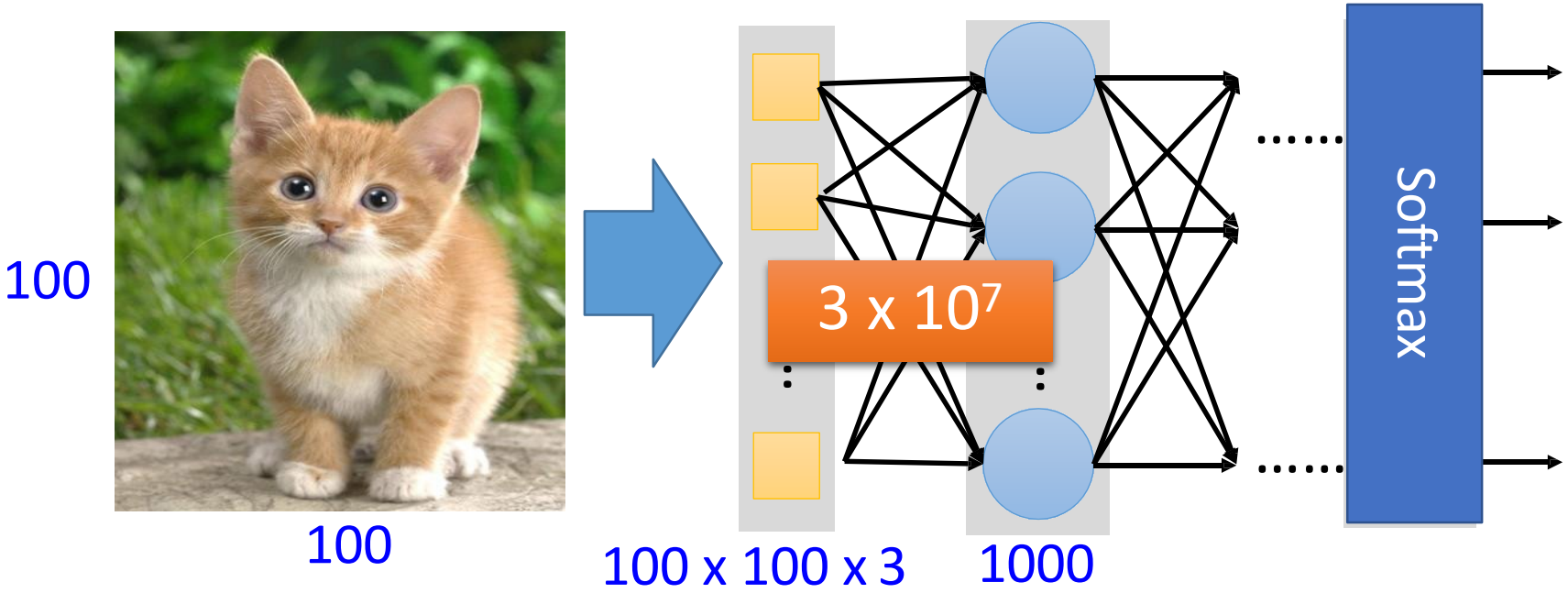
32x32x3 -> 3072 x 1



An Interactive Node-Link Visualization of Convolutional Neural Networks

NN SVG (alexlenail.me)

- 当处理图像时，全连接网络的第一层将会非常大……



可以通过考虑图像识别的性质来简化全连接网络吗?

神经网络的变体

卷积神经网络(CNN)

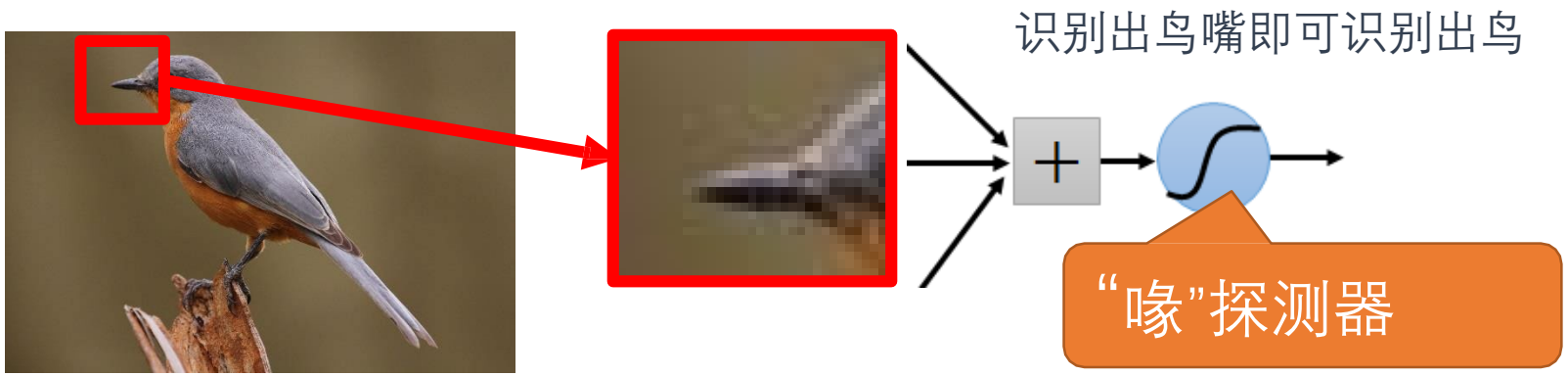
广泛用于
图像处理

为什么 CNN 用于图像识别?

- 一些模式比整张图片小得多

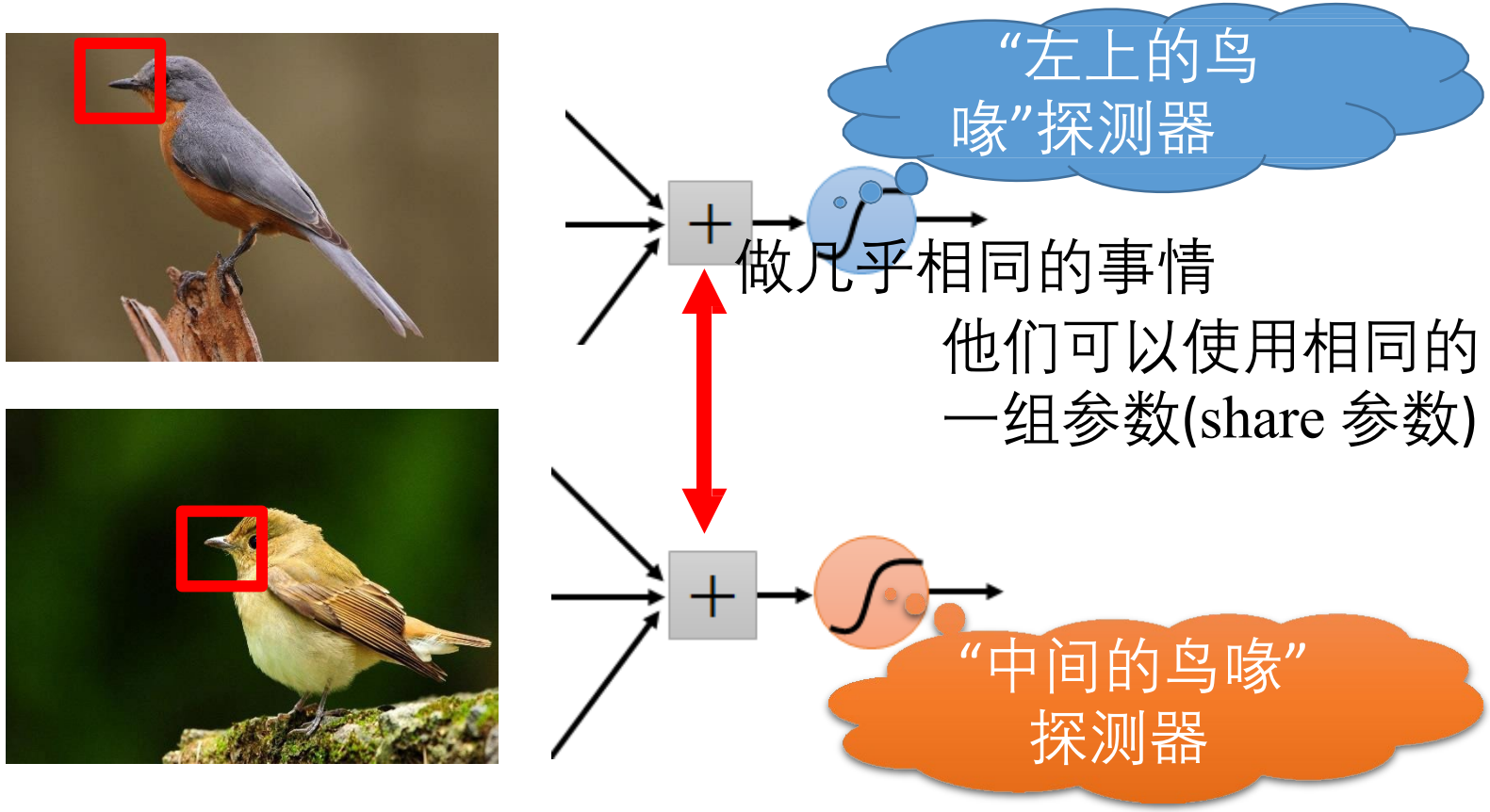
一个神经元不需要看到整个图像去发现模式

通过较少的参数连接到小区域



为什么CNN 用于图像识别?

- 同样的模式可能出现在图像的不同区域



为什么 CNN 用于图像识别?

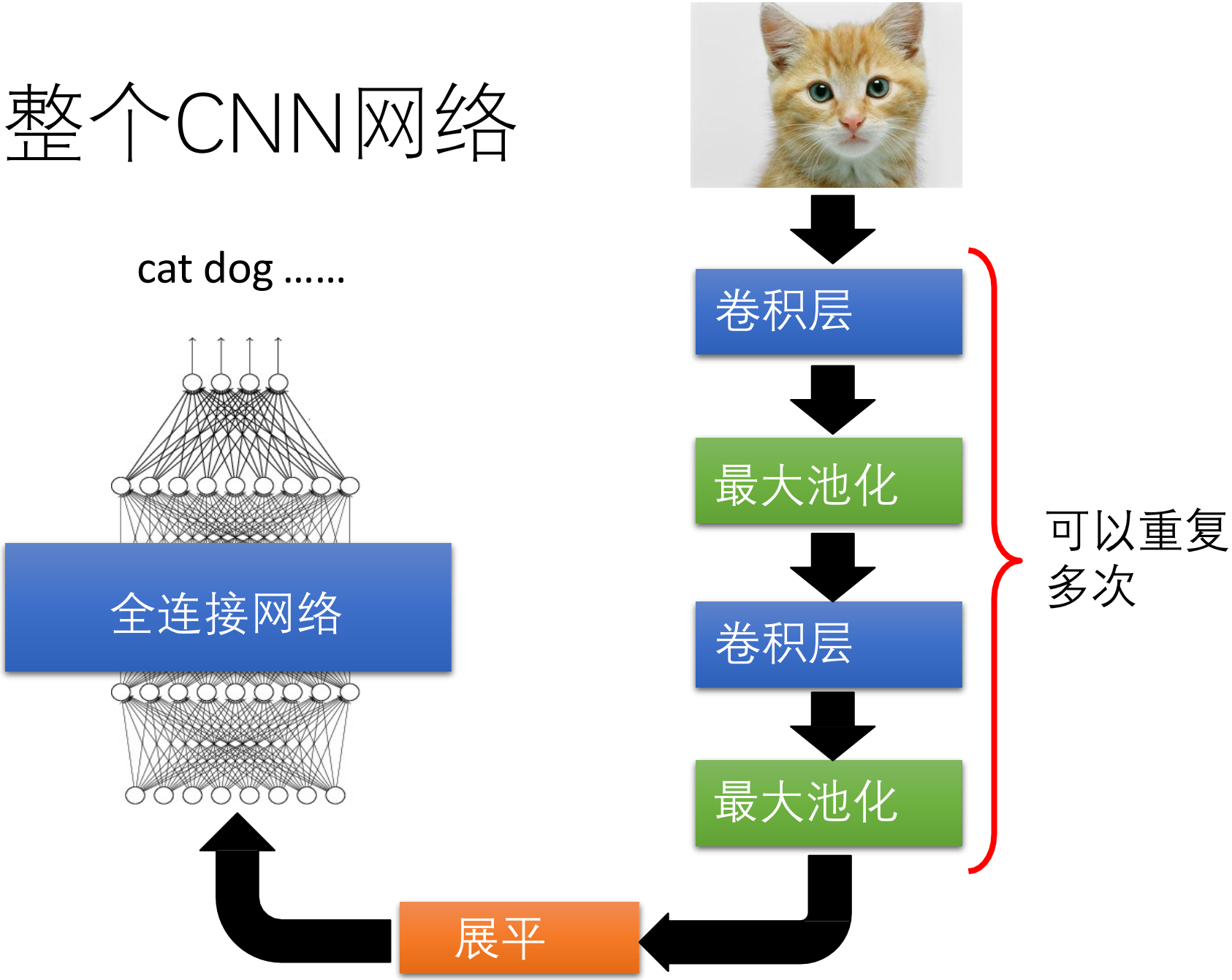
- 对图像进行二次采样不会改变图像中的物体

二次采样

我们可以对图像进行二次采样以使图像变小

更少的参数来处理图像

整个CNN网络



整个CNN网络

属性 1

- 一些模式比整张图片小得多

属性 2

- 同样的模式可能出现在图像的不同区域

属性 3

- 对图像的二次采样不会改变图像中的物体

卷积层

最大池化

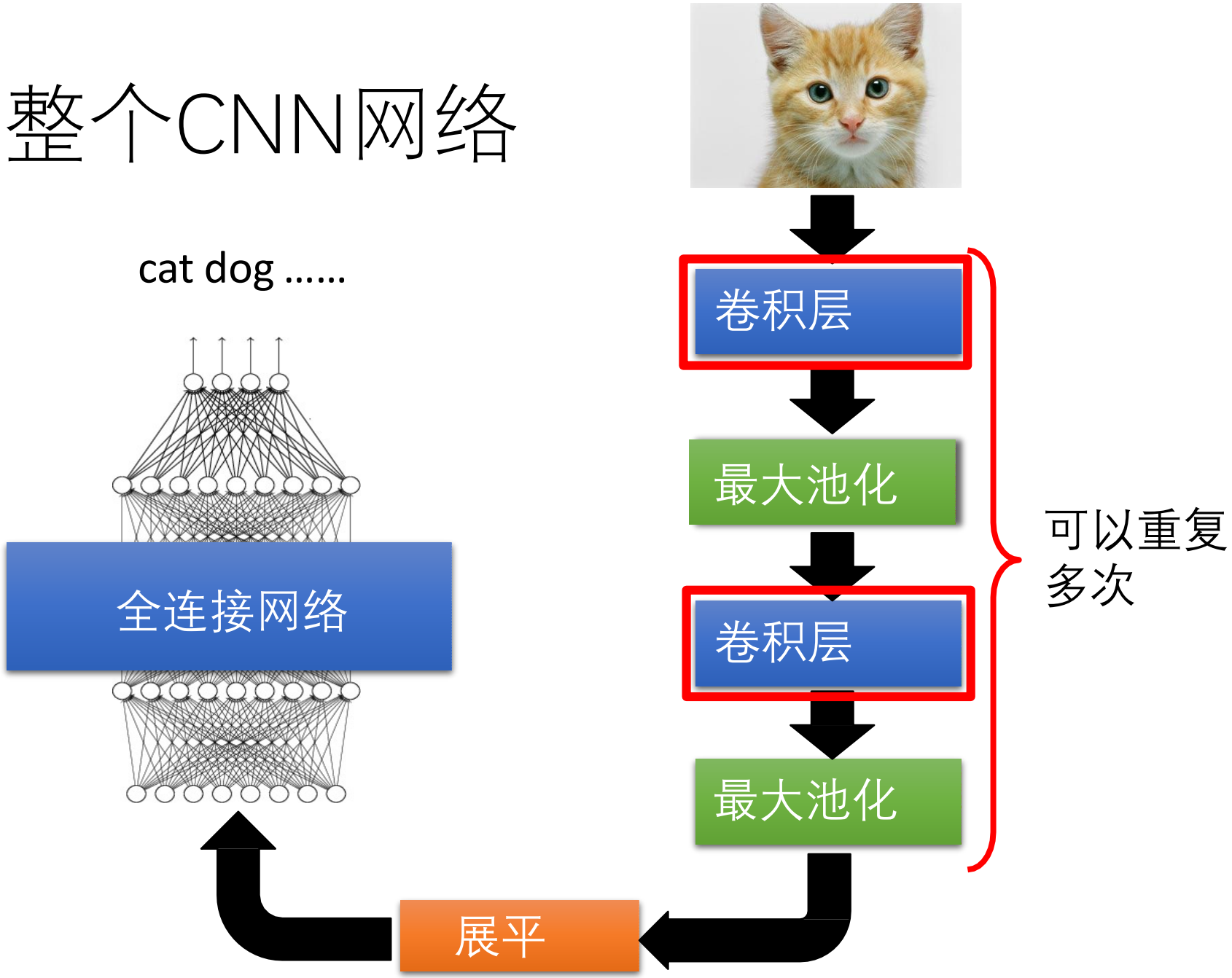
卷积层

最大池化

可以重复多次

Flatten

整个CNN网络



CNN – 卷积

这些都是要学习的参数

Input

1	0	3	1	3	0
3	2	0	1	1	3
2	0	2	0	2	3
1	0	3	0	3	0
3	2	0	0	1	0
2	0	3	0	2	1

6 x 6 image

1	1	0
0	0	1
0	0	0

卷积核 1
矩阵

1	0	0
0	0	0
0	2	0

卷积核 2
矩阵

⋮

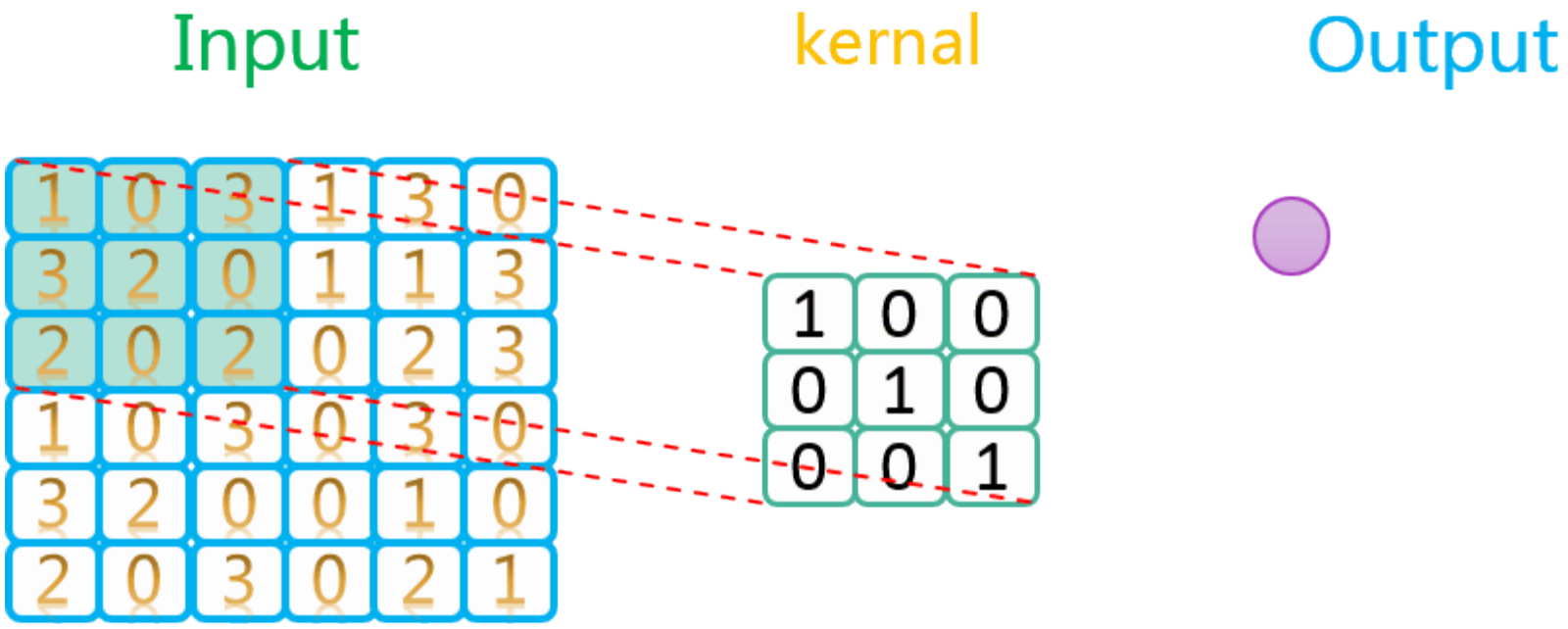
Property 1

每个卷积核探测一个小的模式(3 x 3).

CNN – 卷积

步长=1

$$\text{输出大小} = \frac{\text{输入大小} - \text{卷积核大小}}{\text{步长}} + 1 = \frac{6-3}{1} + 1 = 4$$



CNN –卷积

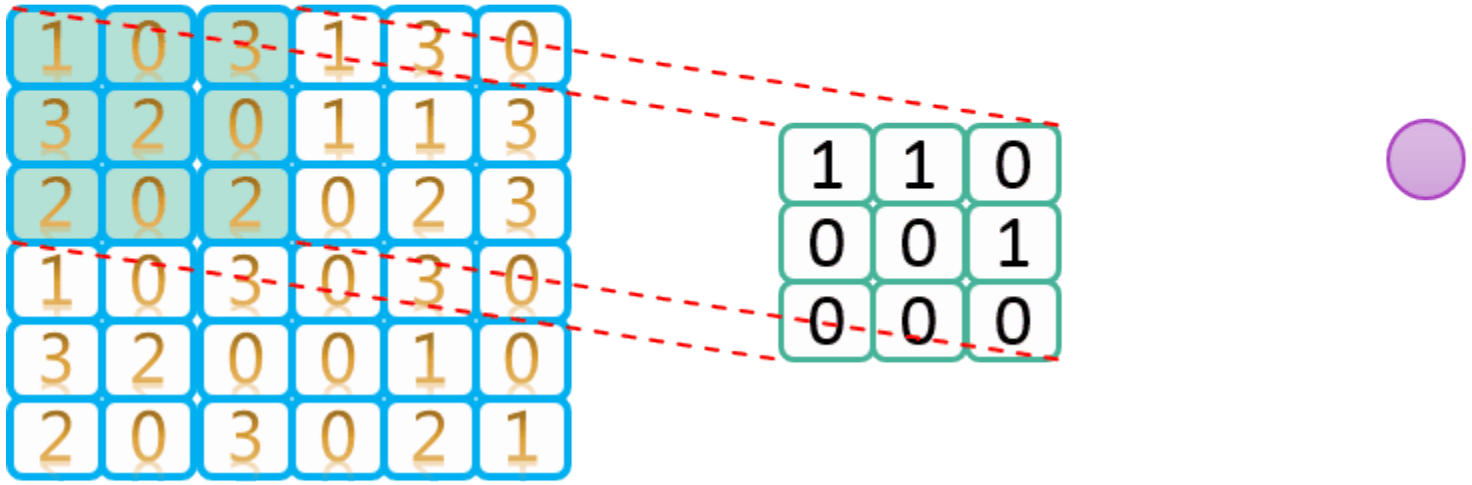
步长=2

$$\text{输出大小} = \frac{\text{输入大小} - \text{卷积核大小}}{\text{步长}} + 1 = \frac{6-3}{2} + 1 = 2$$

Input

kernal

Output



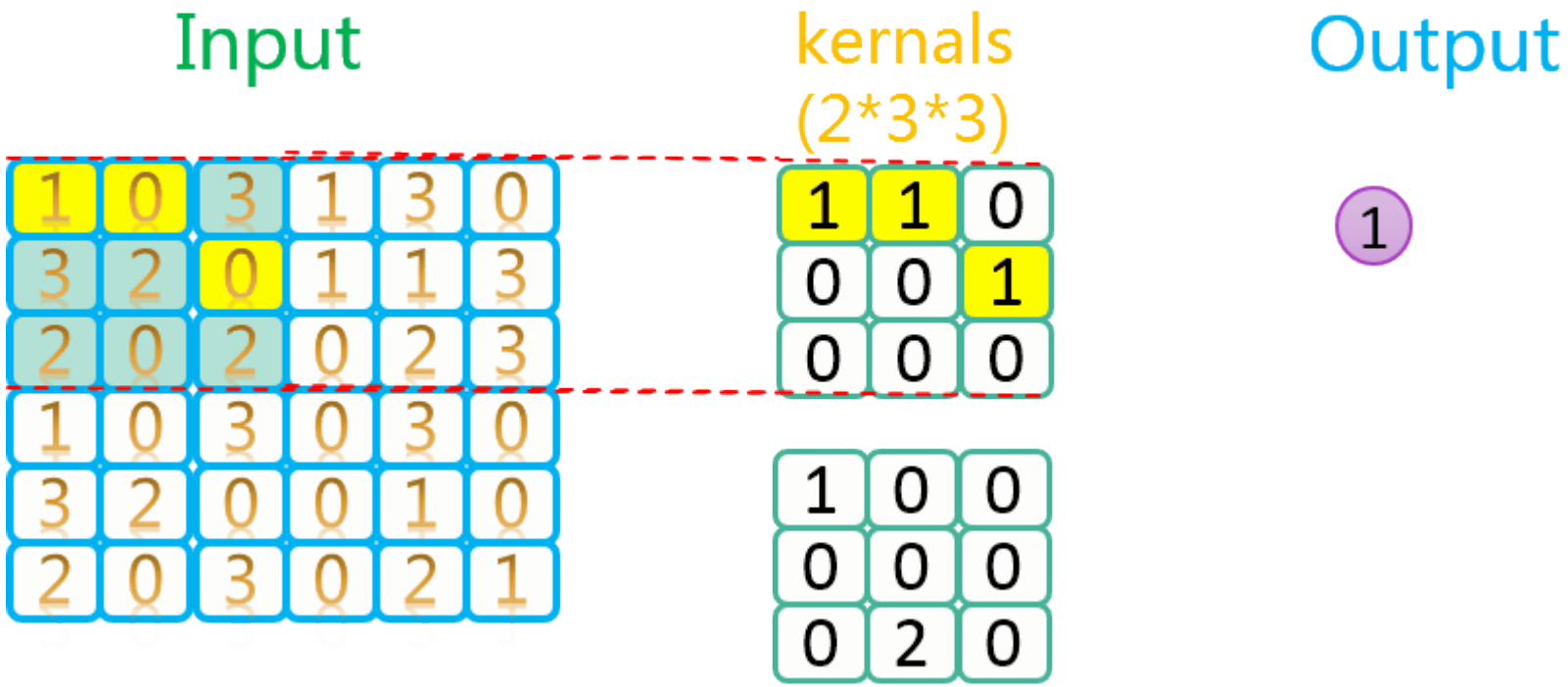
6 x 6 image

我们在下面设置步长=1

CNN –卷积

对于每一个卷积核做同样的事

步长=2



CNN –卷积

步长=1

$$\text{输出大小} = \frac{\text{输入大小}+\text{填充}*2-\text{卷积核大小}}{\text{步长}} + 1 = \frac{6+2*1-3}{1} + 1 = 6$$

填充

0	0	0	0	0	0	0	0
0	3	1	0	2	5	6	0
0	4	2	1	1	4	7	0
0	5	4	0	0	1	2	0
0	1	2	2	1	3	4	0
0	6	3	1	0	5	2	0
0	3	1	0	1	3	3	0
0	0	0	0	0	0	0	0

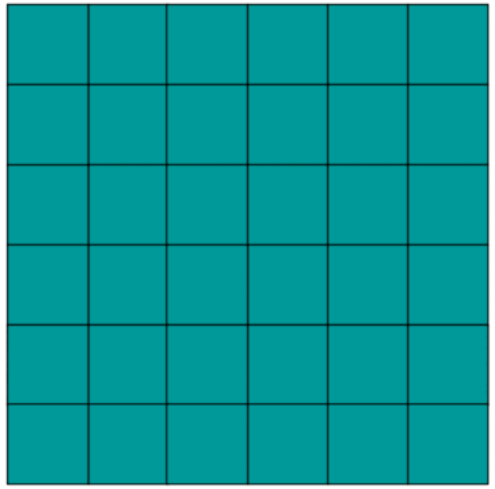
input

*

-1	0	1
-1	0	1
-1	0	1

kernel

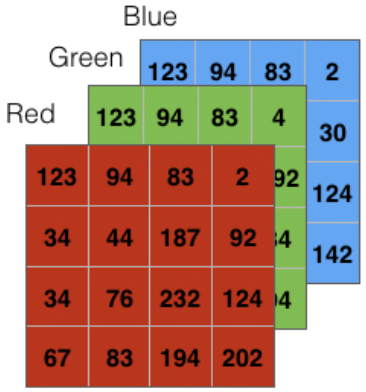
=

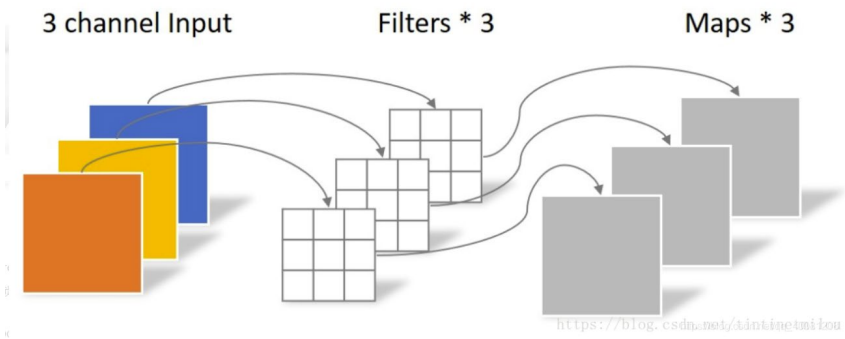


output

CNN – 彩色图像

=





pad

pad

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	123	94	83	2	0	0
0	0	34	44	187	92	0	0
0	0	34	76	232	124	0	0
0	0	67	83	194	202	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

pad

pad

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	123	94	2	4	0	0
0	0	11	3	22	192	0	0
0	0	12	4	23	34	0	0
0	0	194	83	12	94	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

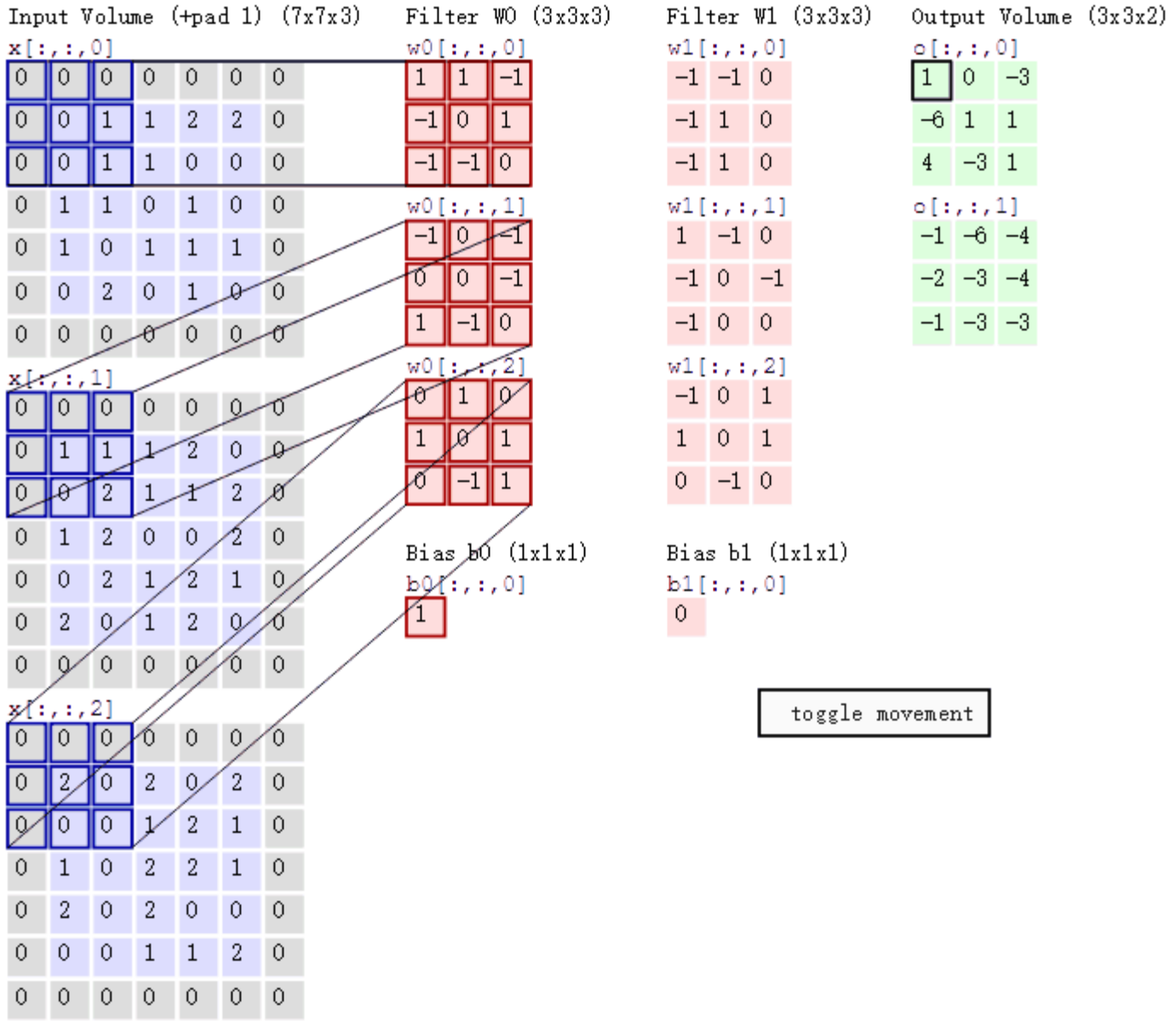
pad

pad

0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	123	94	83	2	0	0
0	0	34	44	37	30	0	0
0	0	34	114	234	124	0	0
0	0	49	18	204	142	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

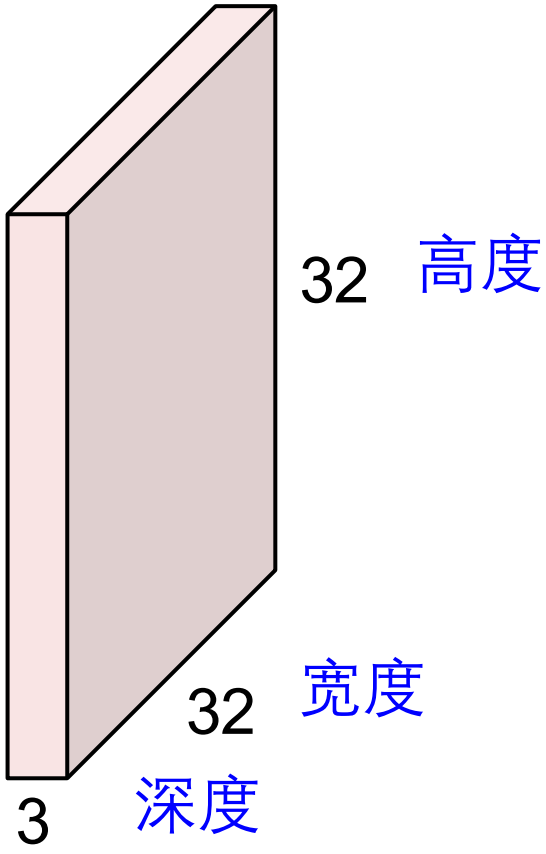
2DConv on RGB image

CNN – 彩色图像


Input Volume (+pad 1) (7x7x3)
x[:,:,0]
0 0 0 0 0 0 0
0 0 1 1 2 2 0
0 0 1 1 0 0 0
0 1 1 0 1 0 0
0 1 0 1 1 1 0
0 0 2 0 1 0 0
0 0 0 0 0 0 0
x[:,:,1]
0 0 0 0 0 0 0
0 1 1 1 2 0 0
0 0 2 1 1 2 0
0 1 2 0 0 2 0
0 0 2 1 2 1 0
0 2 0 1 2 0 0
0 0 0 0 0 0 0
x[:,:,2]
0 0 0 0 0 0 0
0 2 0 2 0 2 0
0 0 0 1 2 1 0
0 1 0 2 2 1 0
0 2 0 2 0 0 0
0 0 0 1 1 2 0
0 0 0 0 0 0 0
Filter W0 (3x3x3)
w0[:,:,0]
1 1 -1
-1 0 1
-1 -1 0
w0[:,:,1]
-1 0 -1
0 0 -1
1 -1 0
w0[:,:,2]
0 1 0
1 0 1
0 -1 1
Bias b0 (1x1x1)
b0[:,:,0]
1
Filter W1 (3x3x3)
w1[:,:,0]
-1 -1 0
-1 1 0
-1 1 0
w1[:,:,1]
1 -1 0
-1 0 -1
-1 0 0
w1[:,:,2]
-1 0 1
1 0 1
0 -1 0
Bias b1 (1x1x1)
b1[:,:,0]
0
Output Volume (3x3x2)
o[:,:,0]
1 0 -3
-6 1 1
4 -3 1
o[:,:,1]
-1 -6 -4
-2 -3 -4
-1 -3 -3
toggle movement

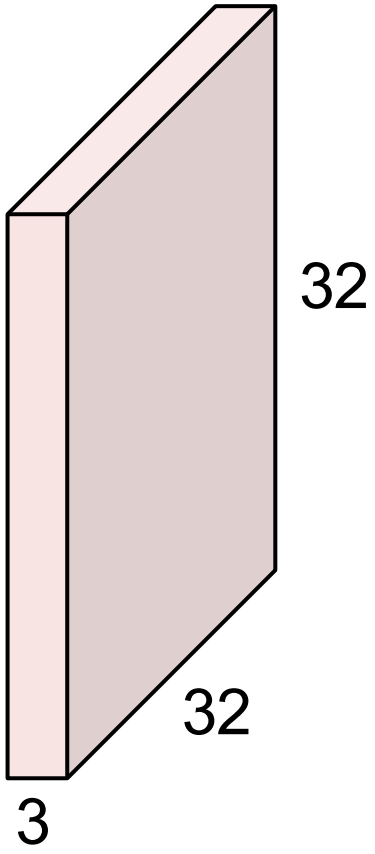
卷积层

32x32x3 image -> 保留空间结构

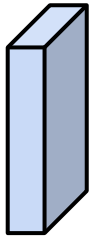


卷积层

32x32x3 图片



5x5x3 卷积核



将卷积核与图像进行卷积，

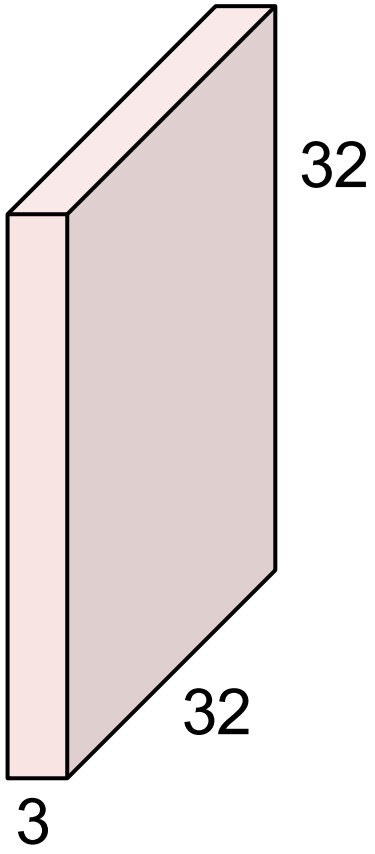
即在空间上滑动图像，计算点积

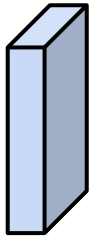
卷积层

卷积核的深度和图片深度相同

32x32x3 图片

5x5x3 卷积核

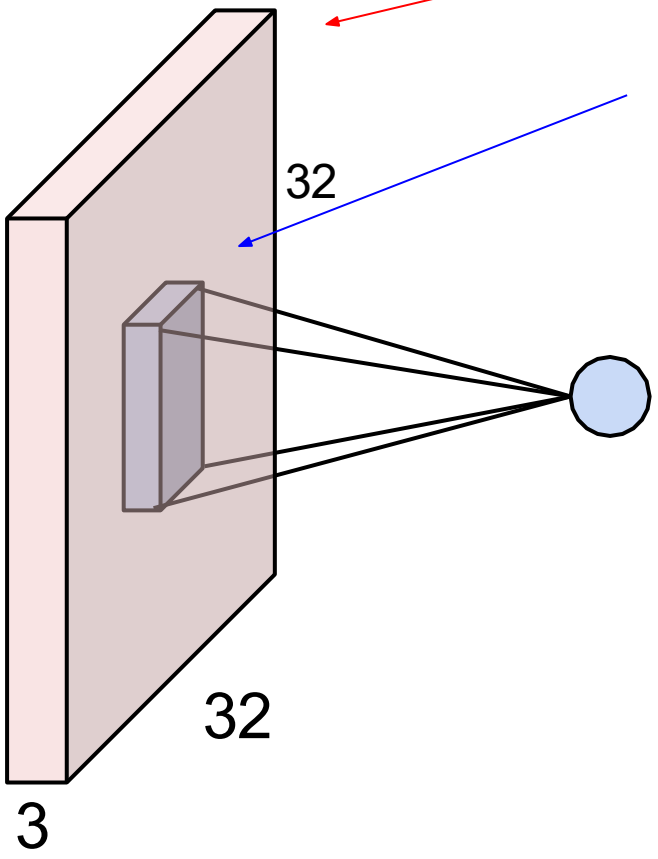




将卷积核与图像进行卷积，

即在空间上滑动图像，计算点积

卷积层



32x32x3 图片

5x5x3 卷积核 w

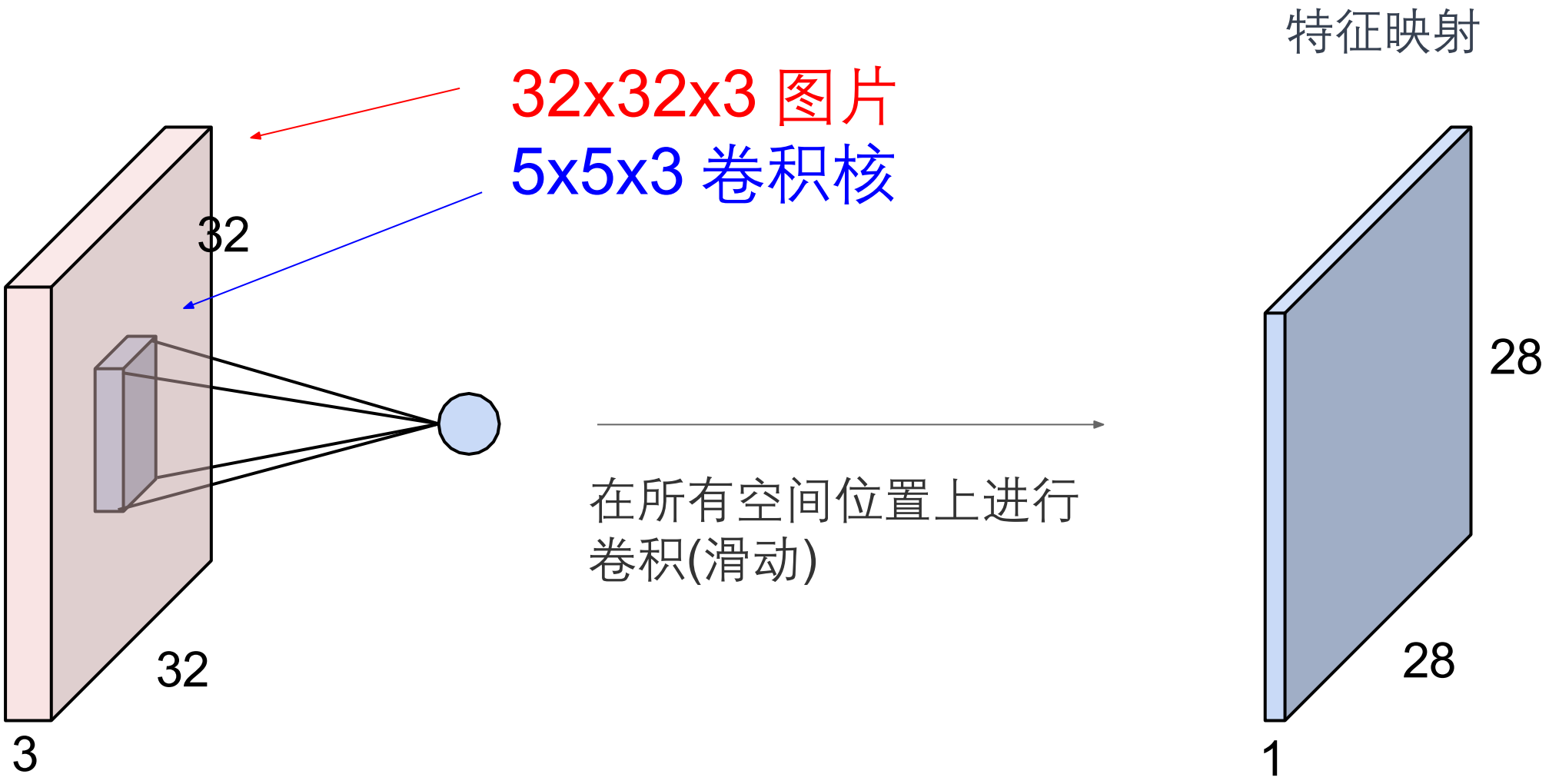
1 number:

卷积核与一小块5x5x3的图像进行点积的结果

(即5*5*3 = 75 维 点积+偏置)

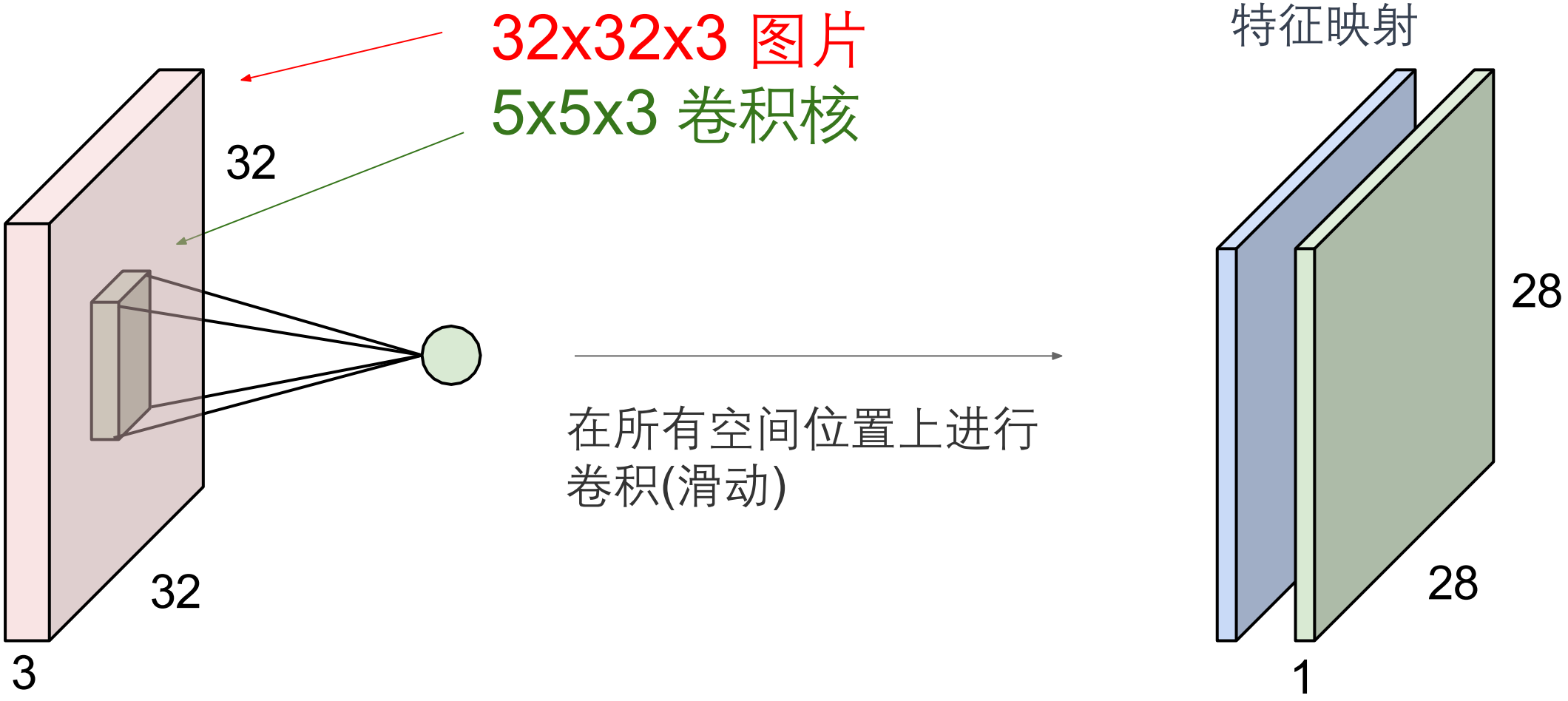
$$w^T x + b$$

卷积层

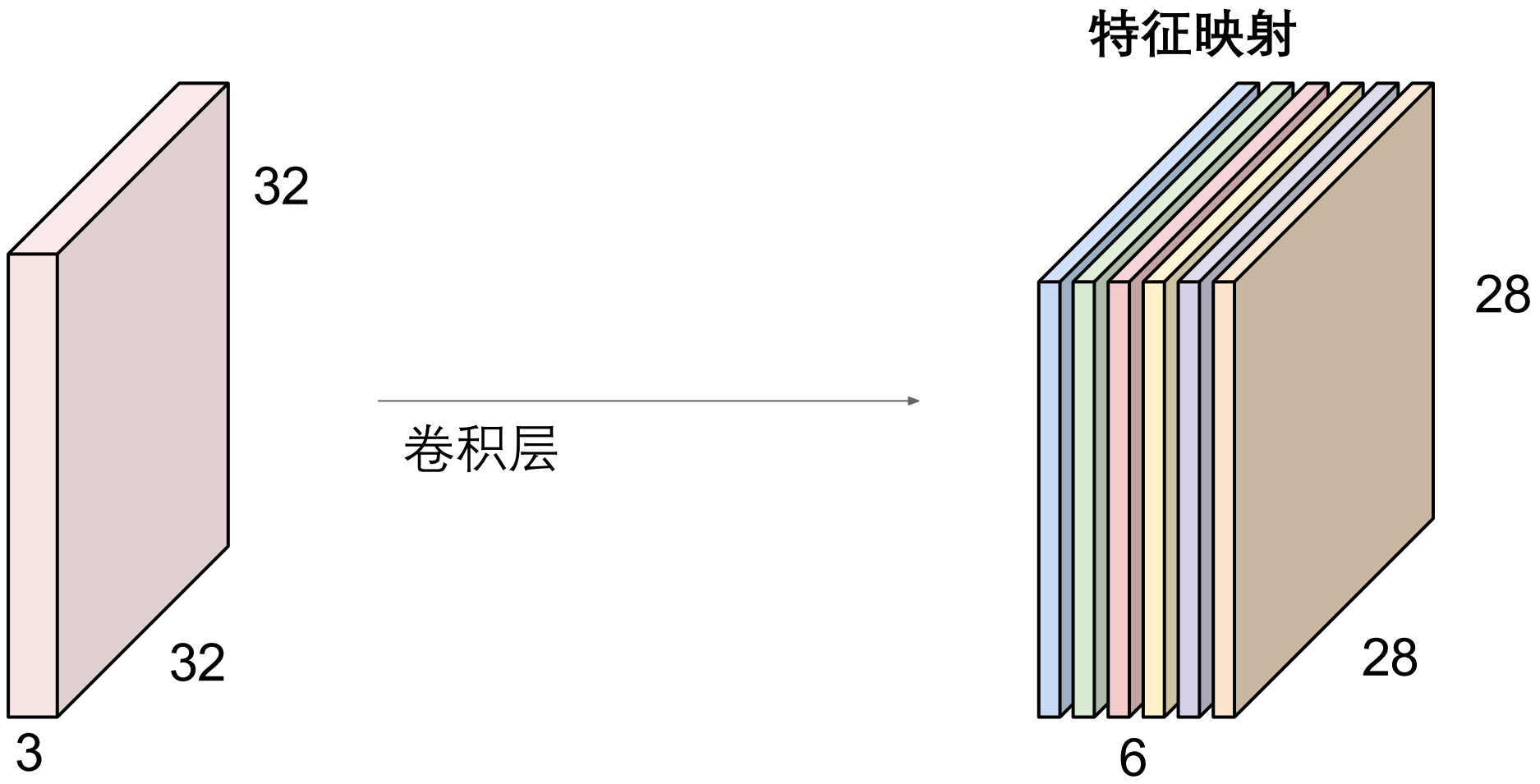


卷积层

考虑第二个绿色卷积核

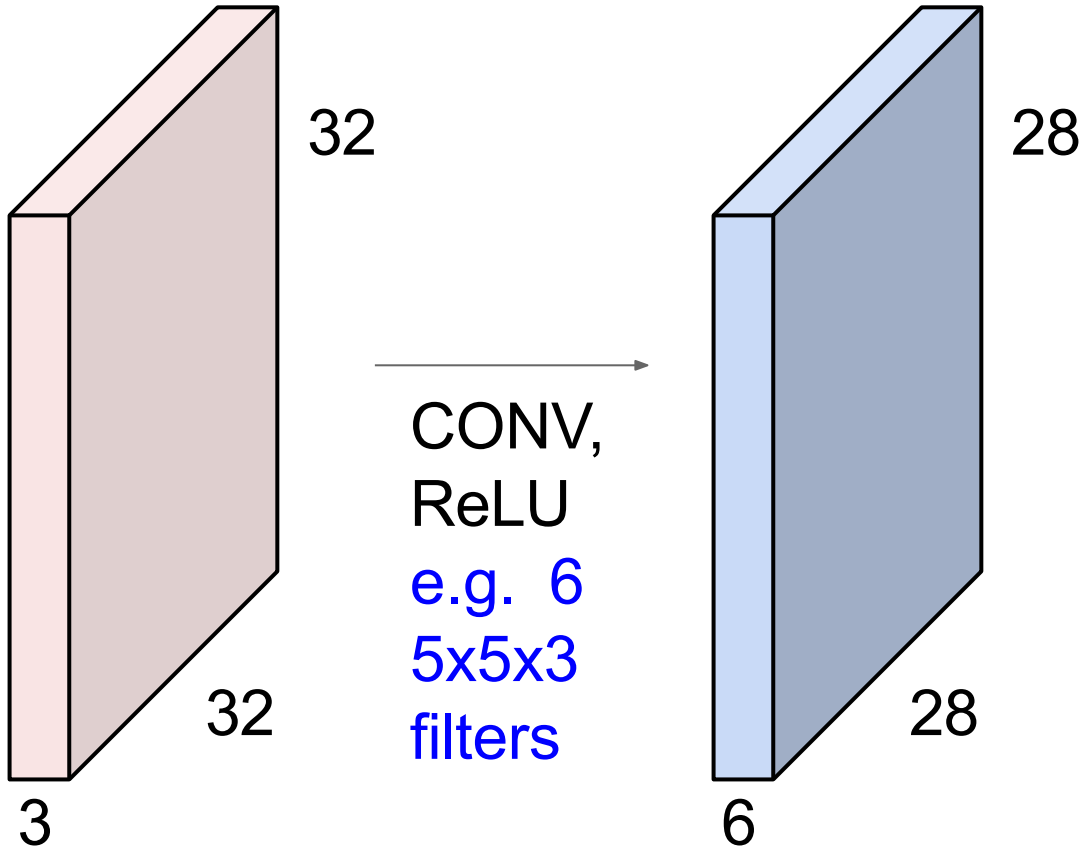


例如，如果我们有6个5x5x3卷积核，我们将得到6个独立的激活映射:

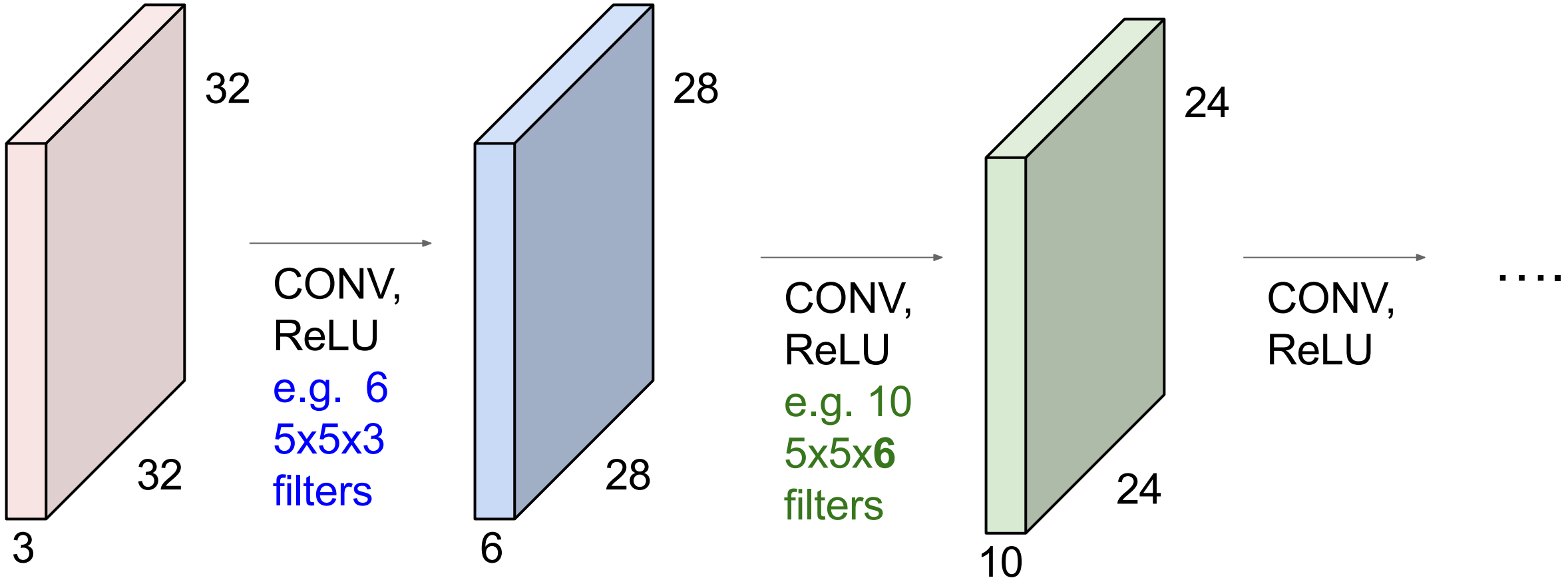


我们将它们堆叠起来，以获得28x28x6尺寸的“新图像”!

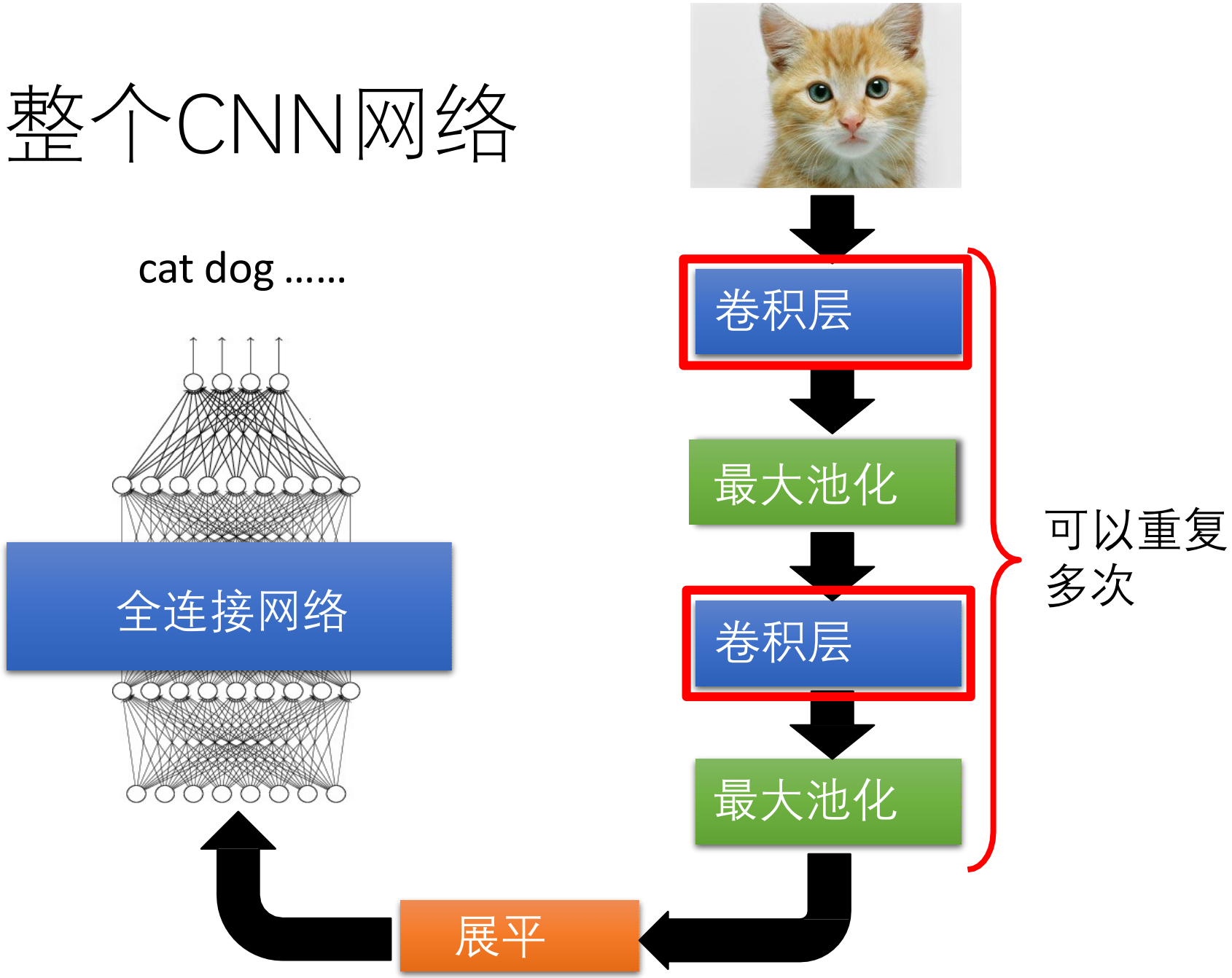
卷积层 = 卷积+激活函数



卷积层 =卷积+激活函数

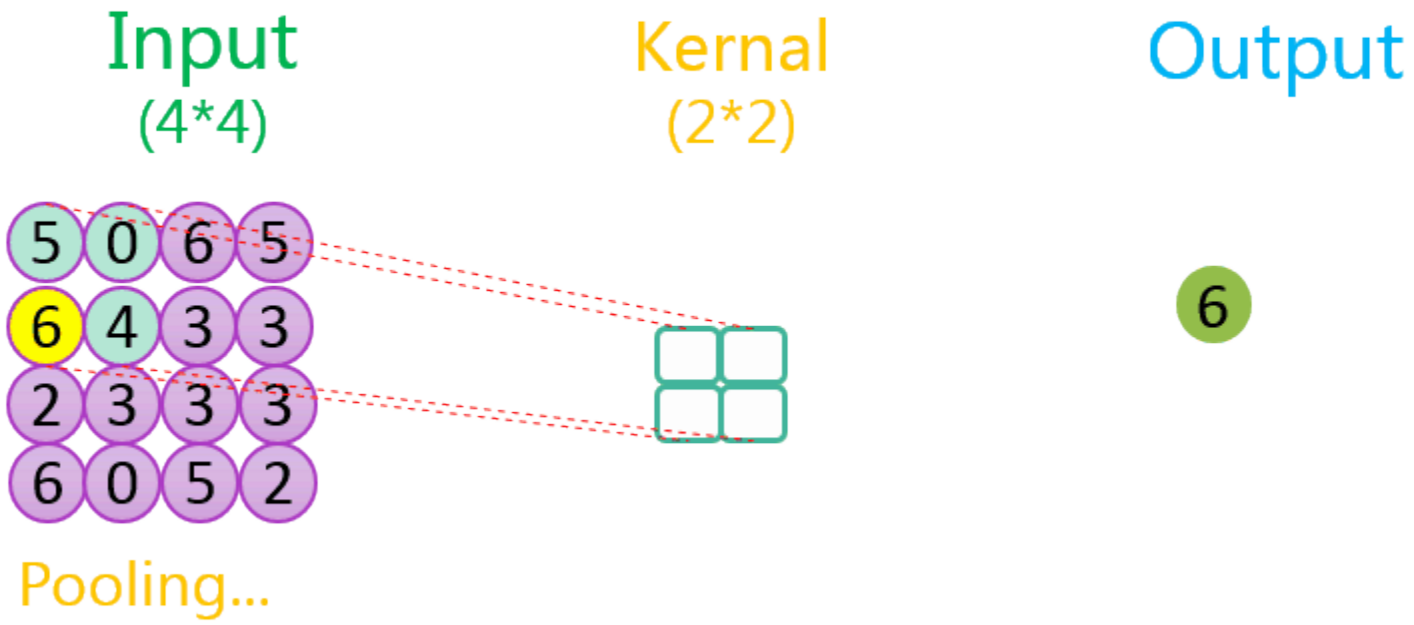

32
32
3
CONV,
ReLU
e.g. 6
5x5x3
filters
28
28
6
CONV,
ReLU
e.g. 10
5x5x6
filters
24
24
10
CONV,
ReLU
....

整个CNN网络

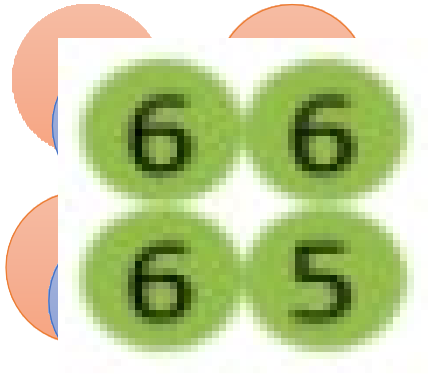


CNN – 最大池化

步长=2

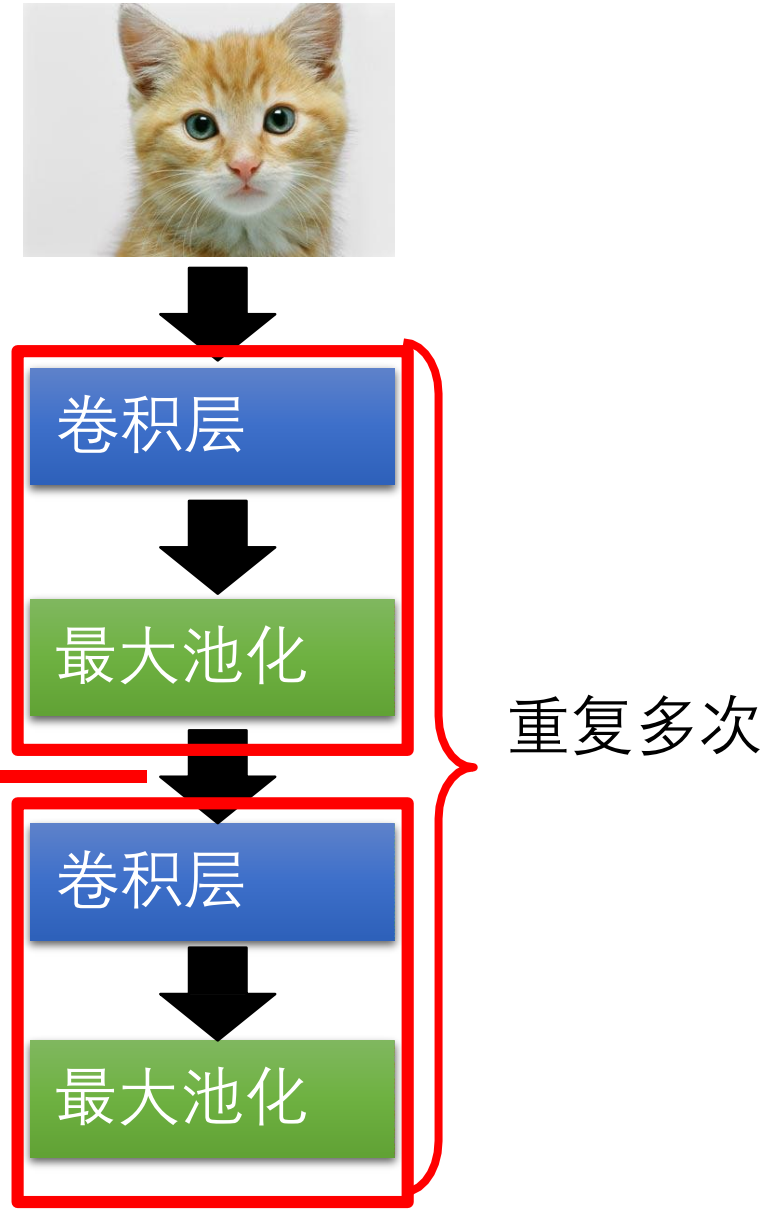


整个CNN网络

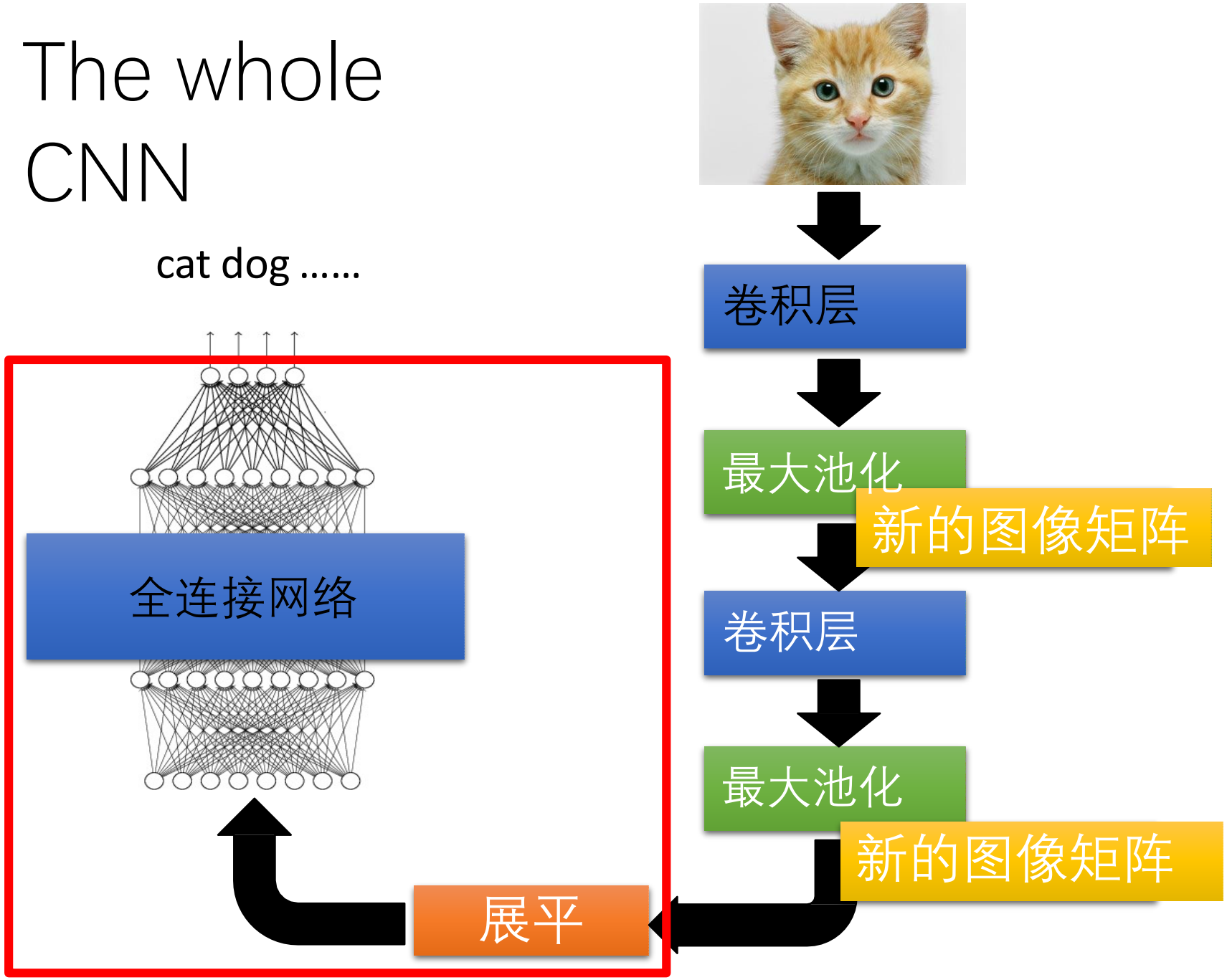


新的图像

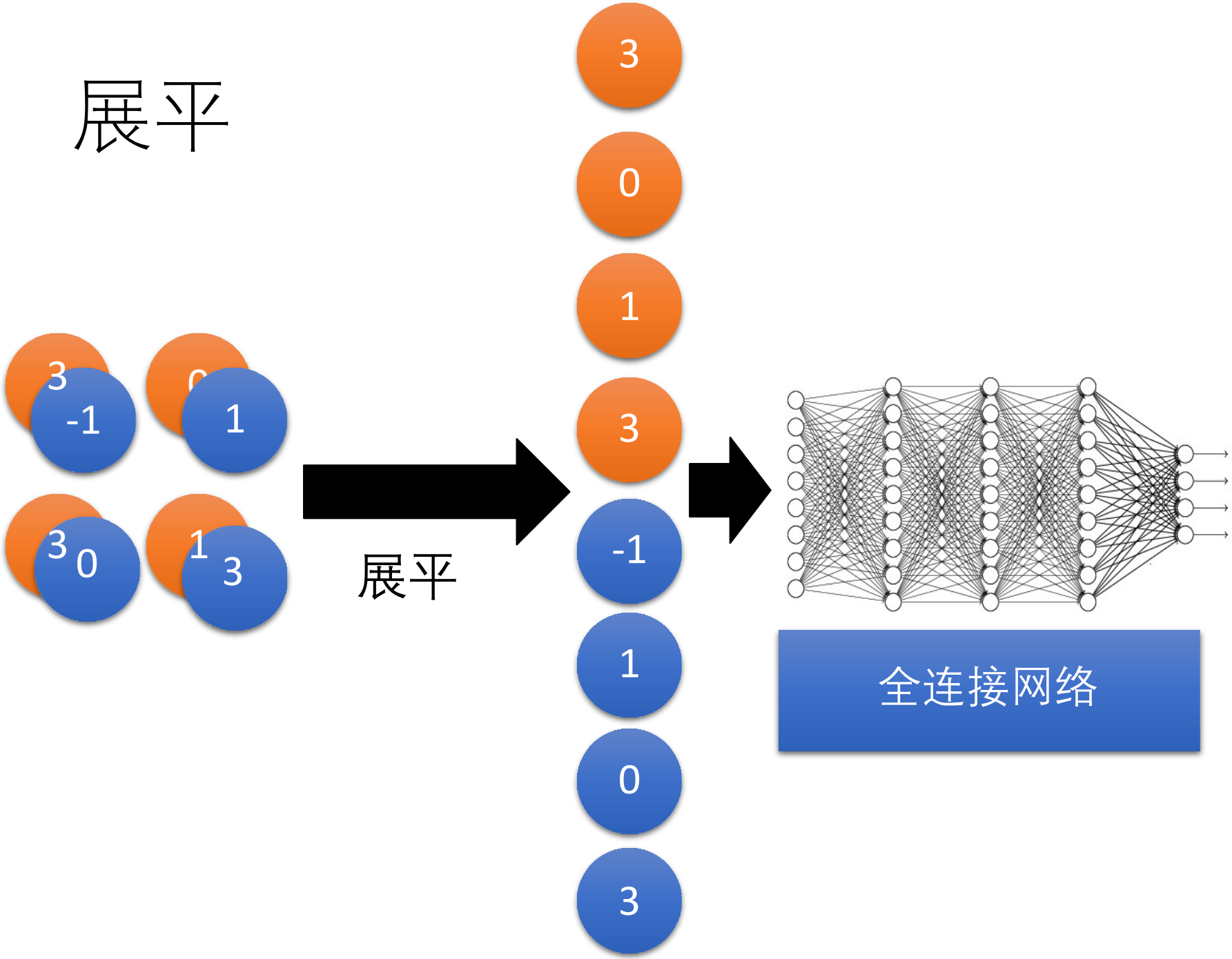
比原始图像更小



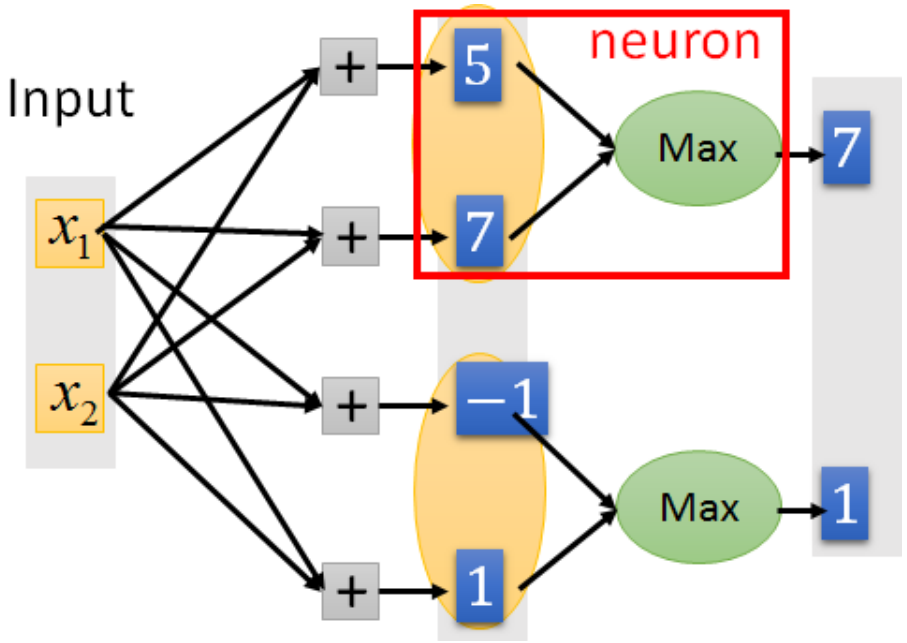
The whole CNN

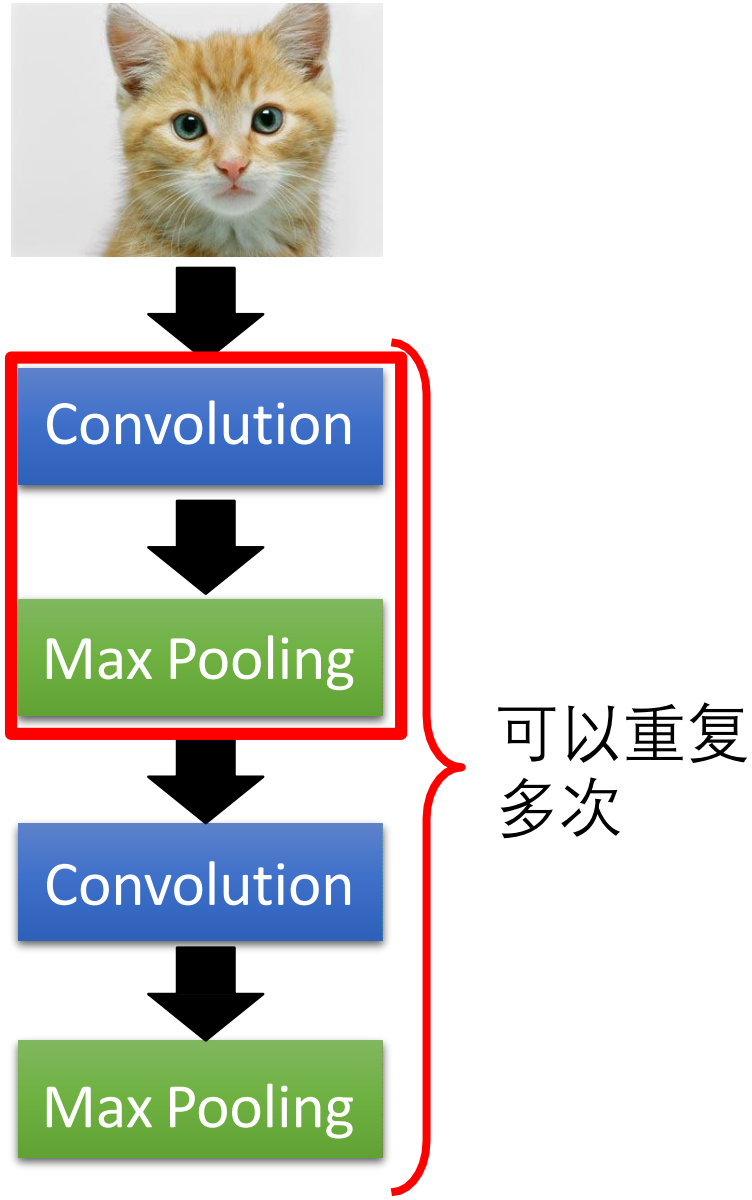


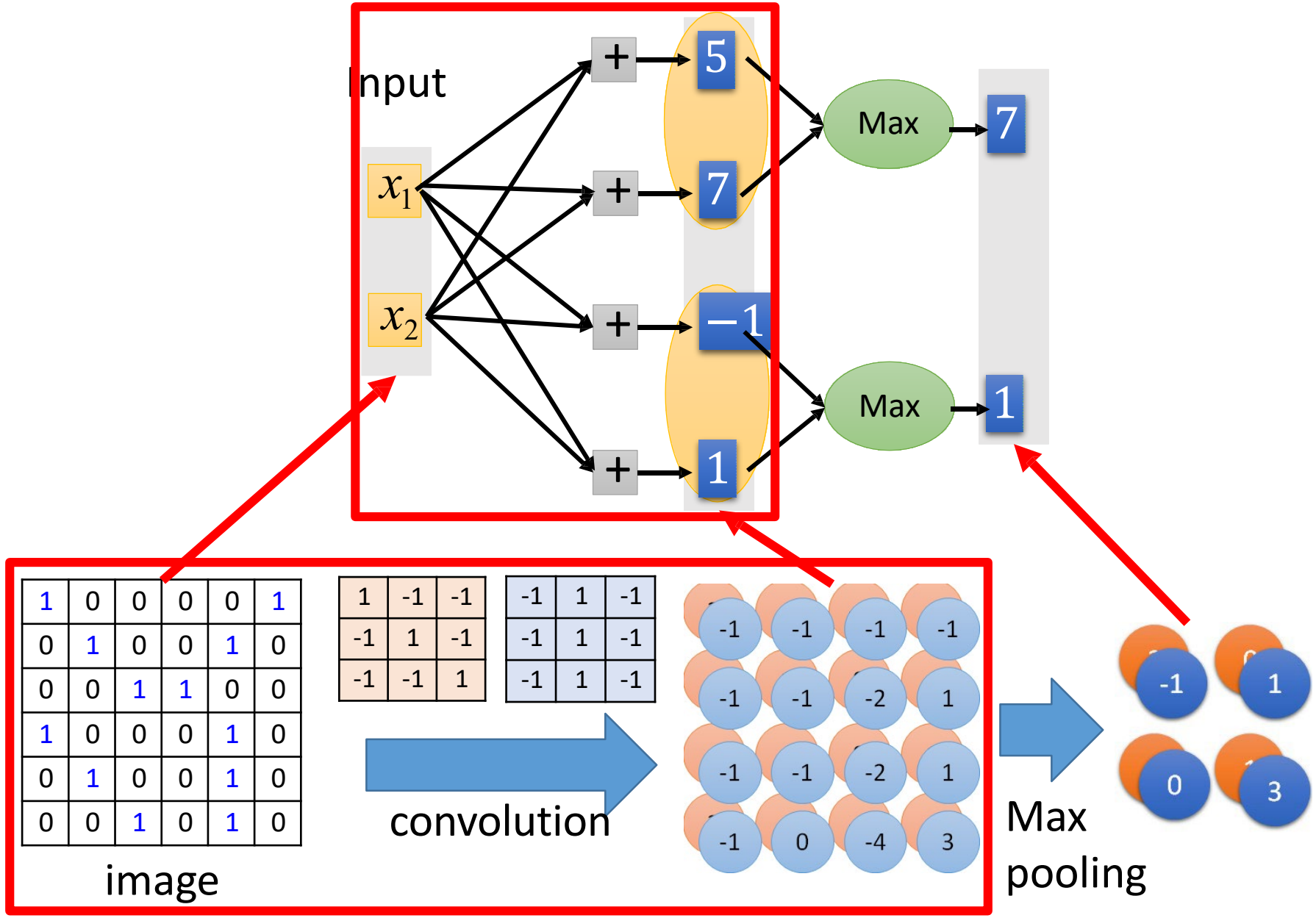
展平



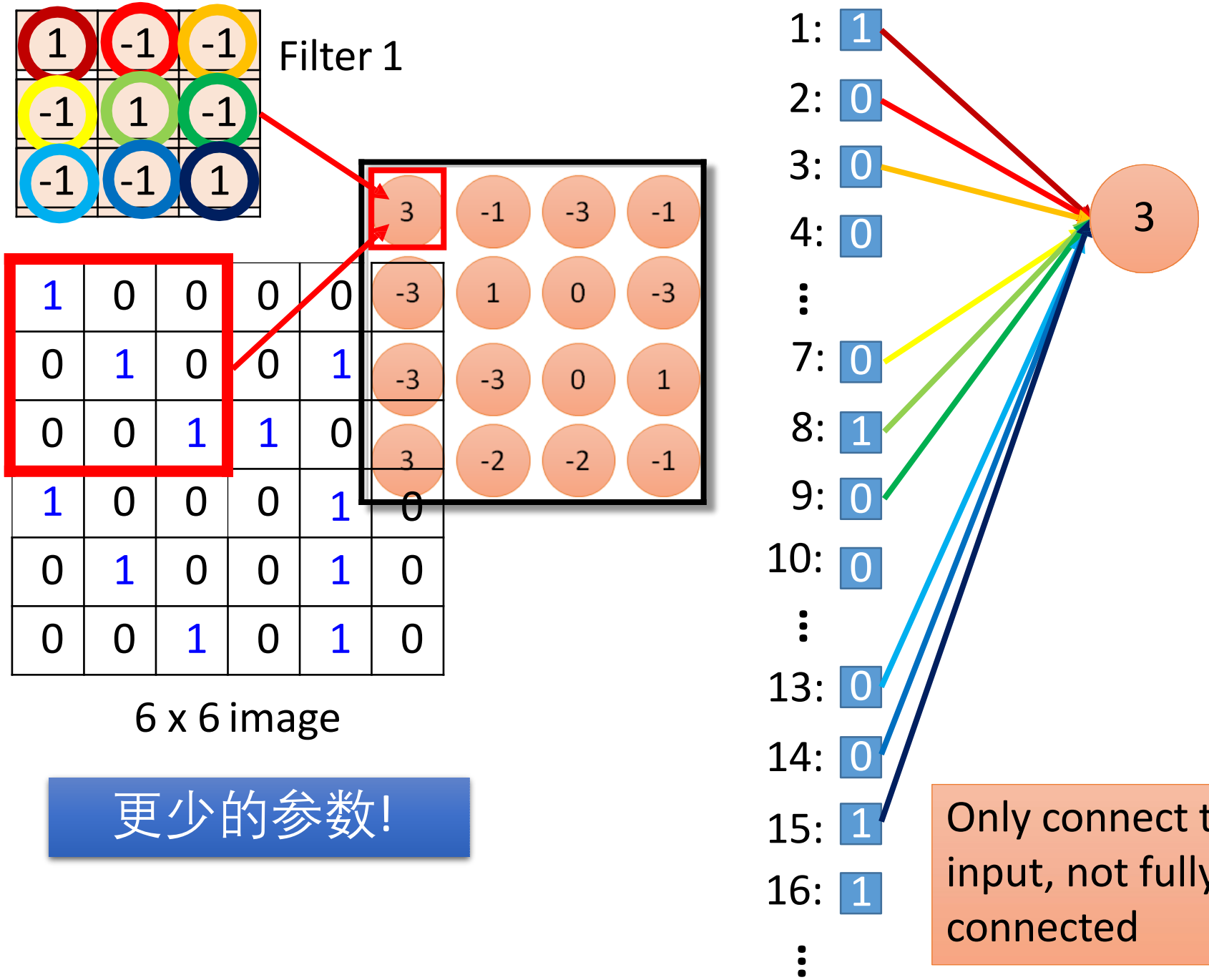
The whole CNN



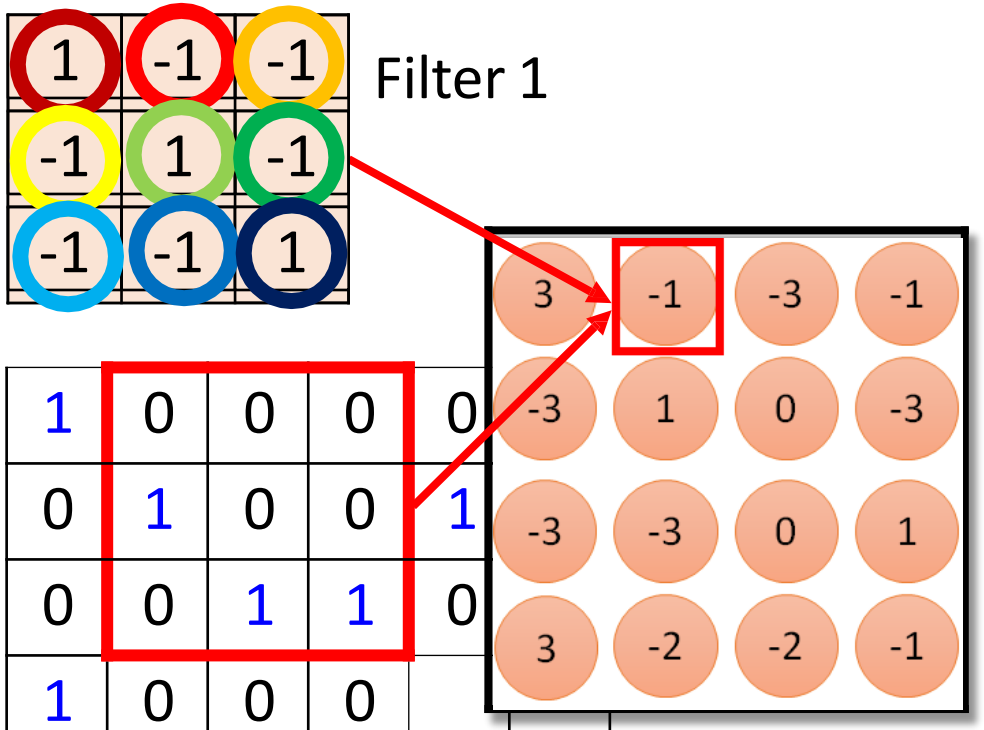




(忽略了卷积后的非线性激活函数)

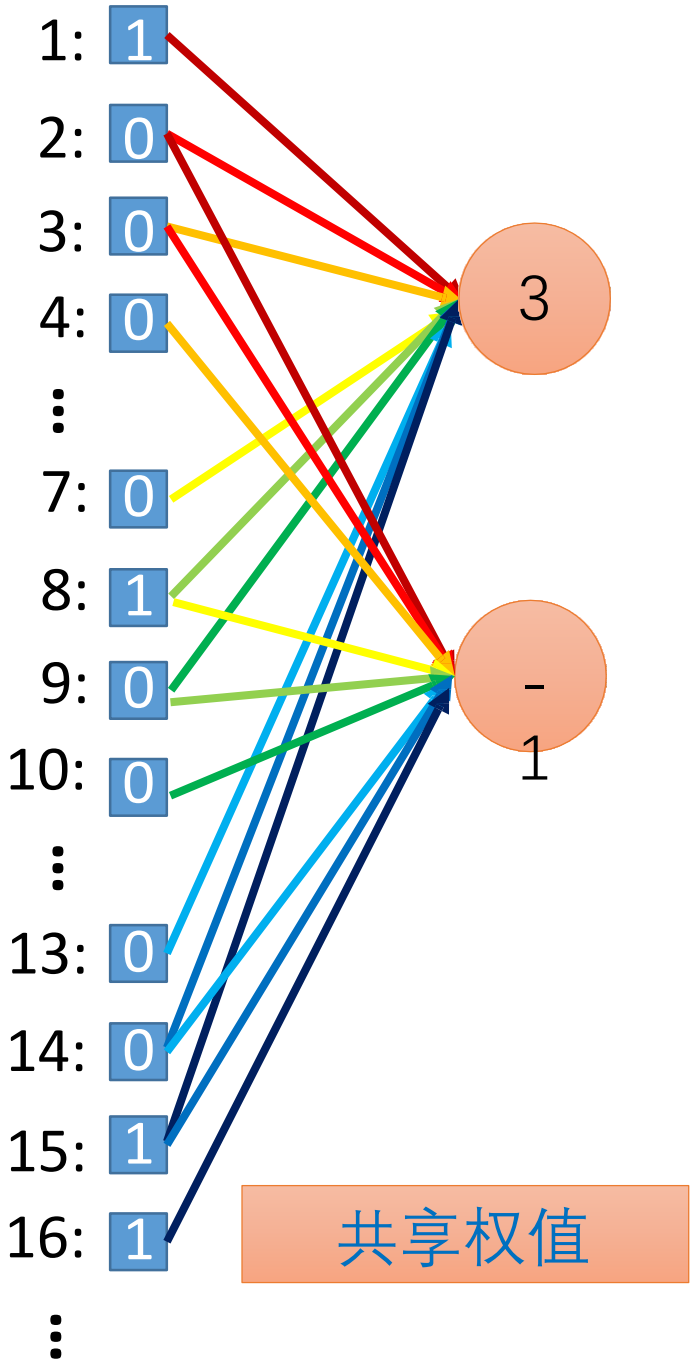

1 -1 -1
-1 1 -1
-1 -1 1
Filter 1
1 0 0 0 0
0 1 0 0 1
0 0 1 1 0
1 0 0 0 1
0 1 0 0 1
0 0 1 0 1
3 -1 -3 -1
-3 1 0 -3
-3 -3 0 1
3 -2 -2 -1
6 x 6 image
1: 1
2: 0
3: 0
4: 0
7: 0
8: 1
9: 0
10: 0
13: 0
14: 0
15: 1
16: 1
3
Only connect to 9 input, not fully connected

更少的参数!

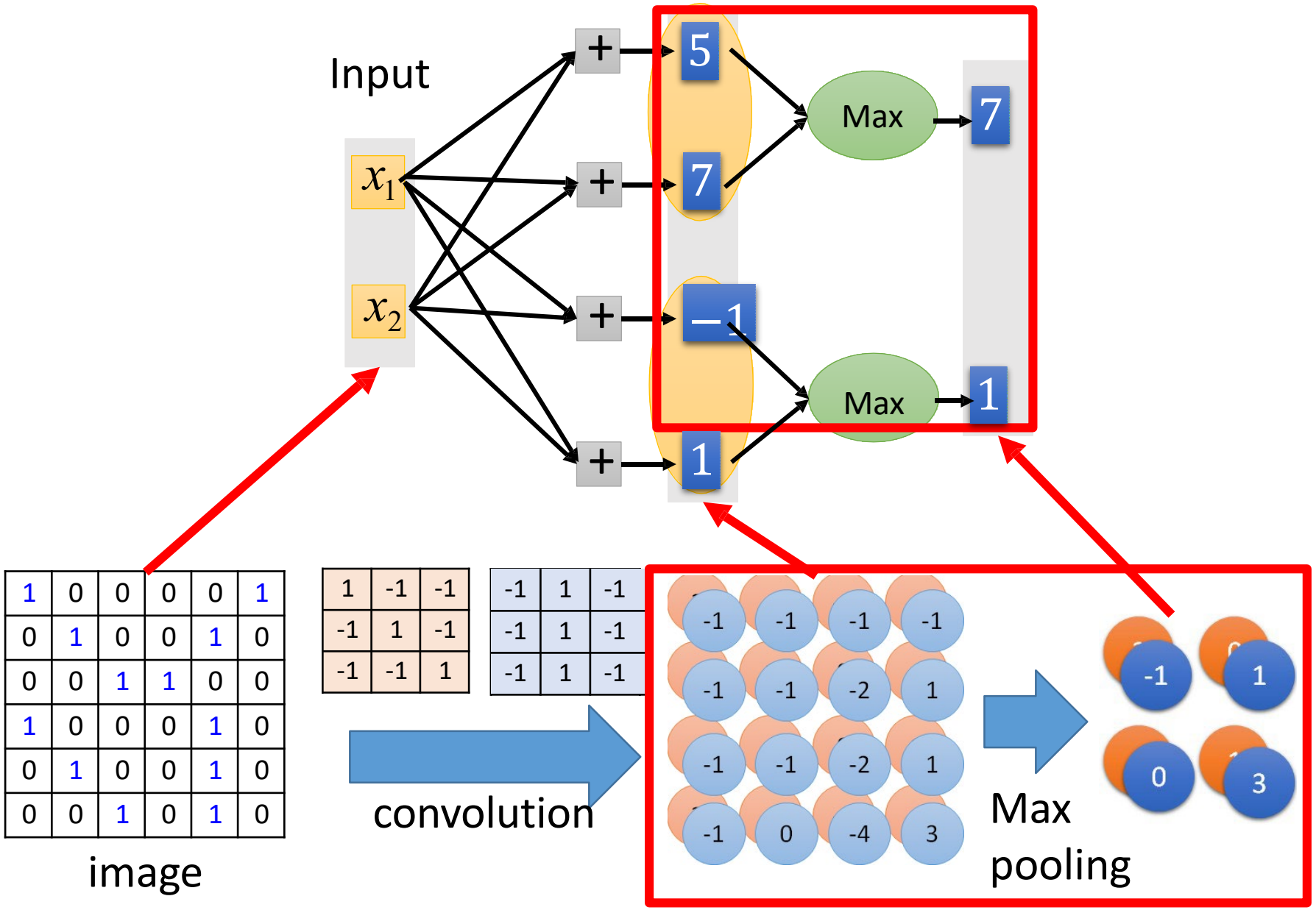


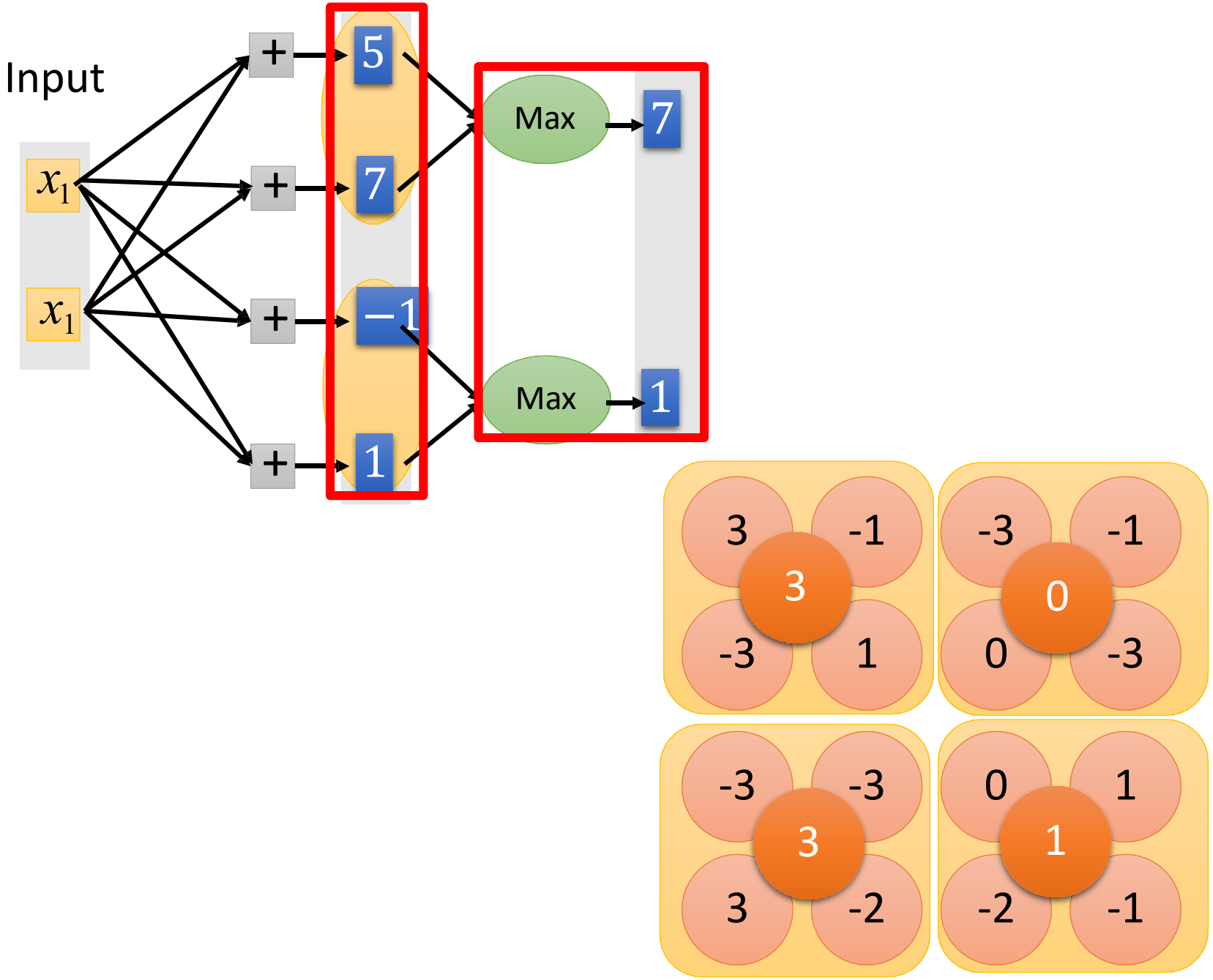
6 x 6 image

更少的参数!



共享权值

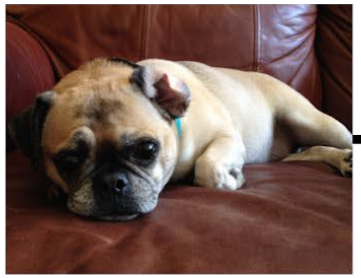

Input
x_1
x_2
+
+
+
+
5
7
−1
1
Max
Max
7
1
1 0 0 0 0 1
0 1 0 0 1 0
0 0 1 1 0 0
1 0 0 0 1 0
0 1 0 0 1 0
0 0 1 0 1 0
image
1 -1 -1
-1 1 -1
-1 -1 1
-1 1 -1
-1 1 -1
-1 1 -1
convolution
-1 -1 -1 -1
-1 -1 -2 1
-1 -1 -2 1
-1 0 -4 3
Max
pooling
-1 1
0 3


Input
x_1
x_1
+
+
+
+
5
7
−1
1
Max
Max
7
1
3
-1
3
-3
1
-3
-1
0
0
-3
-3
-3
3
3
-2
0
1
1
-2
-1

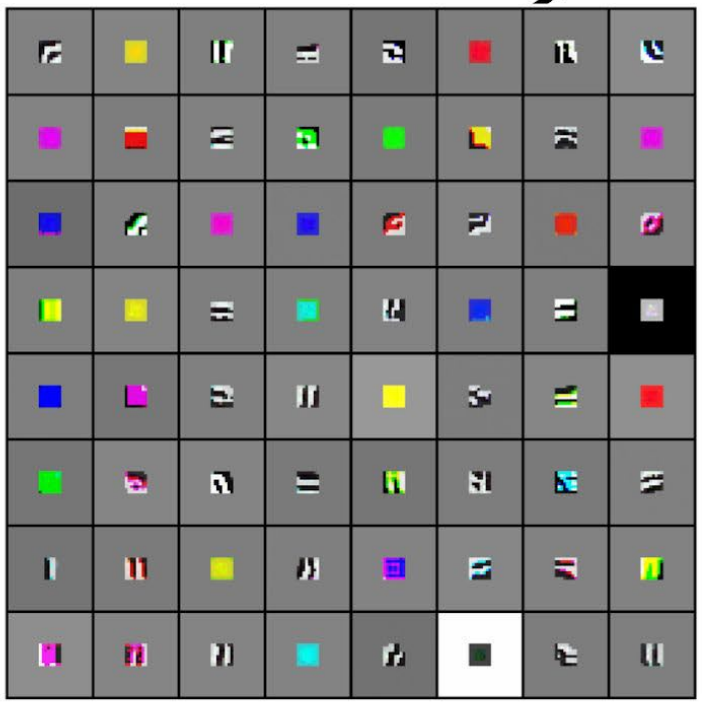
Preview

[Zeiler and Fergus 2013]

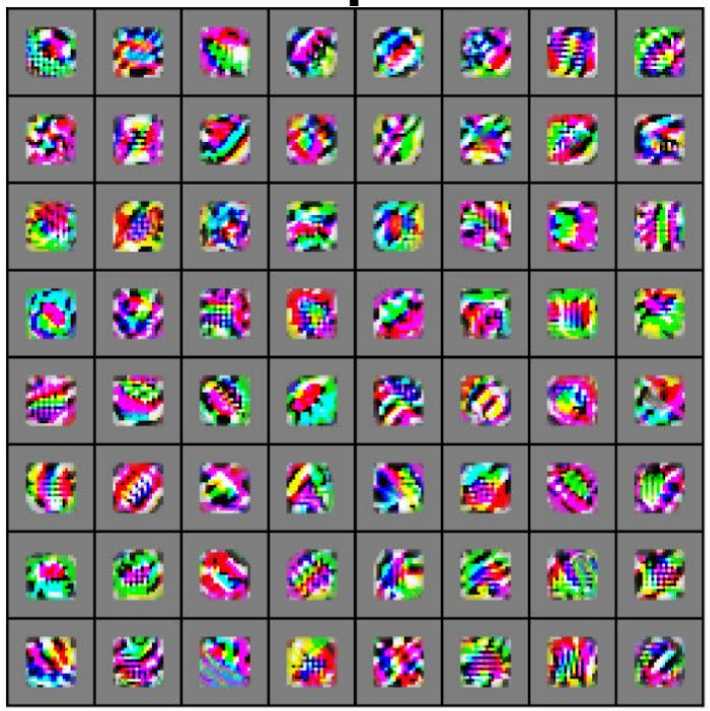
Visualization of VGG-16 by Lane McIntosh. VGG-16 architecture from [Simonyan and Zisserman 2014].



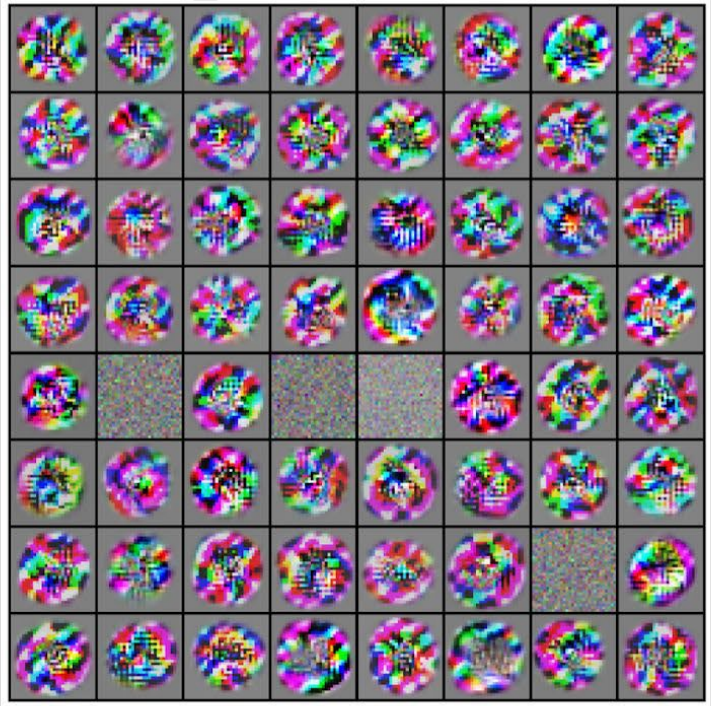
Low-level features → Mid-level features → High-level features → Linearly separable classifier →



VGG-16 Conv1_1

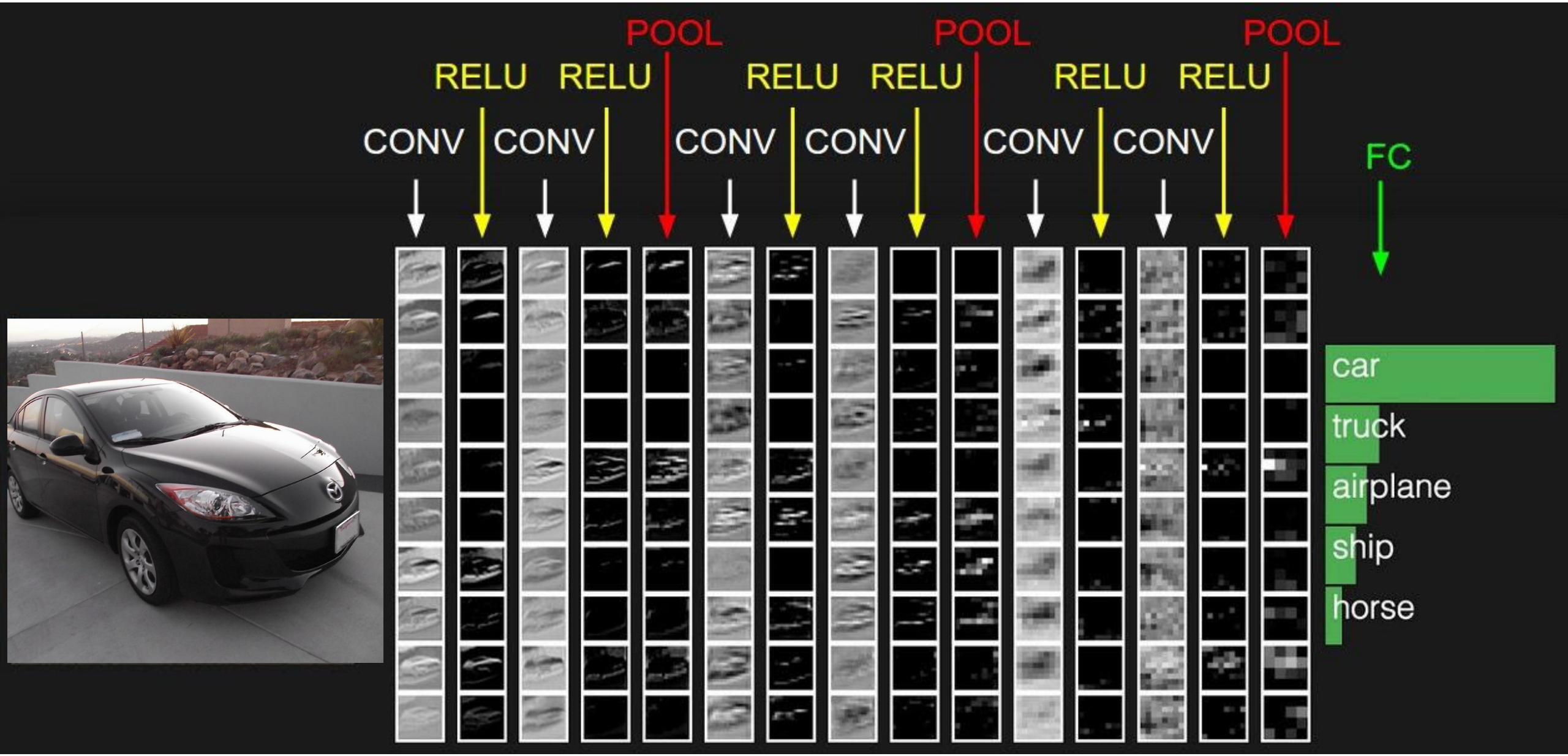


VGG-16 Conv3_2



VGG-16 Conv5_3

preview:


POOL
POOL
POOL
RELU
RELU
RELU
RELU
RELU
RELU
CONV
CONV
CONV
CONV
CONV
CONV
FC
car
truck
airplane
ship
horse